

A Structural Spillover Network Integrating Diebold-Yilmaz and SVAR: Evidence from China's Interbank Market

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Abstract

We propose a structural spillover network framework by integrating the Diebold-Yilmaz network with structural vector autoregressive (SVAR) methodology. This framework is easily implementable and offers a structural interpretation of network spillovers. Applied to China's interbank rate network and monetary policy shocks, the framework reveals that Treasury bond yields, especially at 1- and 10-year maturities, act as central hubs, while the roles of money-market rates are peripheral. Moreover, a multi-step regression strategy reveals a strong concordance between the network topology and the explanatory power of monetary policy shocks for interest rate dynamics, which validates our empirical findings. The proposed framework provides a general approach for mapping shock propagation in macroeconomic and financial systems.

Keywords: Spillover Network; SVAR; Shock Transmission; China's Monetary Policy

JEL Classification: C58; E44; G12

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1 Introduction

In SVAR-based macroeconomic and financial analysis, standard tools such as impulse response functions (IRFs) and forecast error variance decompositions (FEVDs) are widely employed to assess the effects of structural shocks on the variables of interest (Kilian and Lütkepohl, 2017). The total effect of a shock on any variable, however, is mediated by inter-variable linkages, meaning that shock-propagation reflects both direct transmission and indirect spillovers via the system’s network structure (Billio et al., 2012; Diebold and Yilmaz, 2012, 2014). While IRFs and FEVDs characterize how shocks propagate through individual variables over time (transmission), they are less suited to examining how shocks propagate across multiple variables within a system (spillovers).

The network-based connectedness framework developed by Diebold and Yilmaz (2012, 2014) (hereafter, D&Y network) provides an approach for measuring the spillover effects. However, the D&Y network remains a reduced-form construct: it is built on generalized forecast error variance decompositions (GFEVDs), which measures how reduced-form innovations in one variable helps to reduce the mean square forecast errors (MSFE) in others without distinguishing the underlying structural sources. Consequently, each edge of the D&Y network mix all innovations, limiting its economic interpretation. For instance, it is not possible to isolate contributions from “common shocks” such as monetary, growth, or risk-premium shocks (Cieslak and Pang, 2021), nor to conduct shock-specific policy counterfactuals.

To address these limitations, we propose a novel framework that integrates the D&Y network with the SVAR methodology to construct a structural shock spillover network. From an SVAR perspective, reduced-form innovations are linear combinations of structural shocks. Therefore, the information contained in one variable’s reduced-form innovation for forecasting another variable, as measured by the GFEVDs, ultimately stems from the contemporaneous causal effects of underlying structural shocks. The connectivity and weights within the D&Y network thus provide an empirical mapping of the aggregate pathways

through which structural shocks propagate. For example, if variable j is affected predominantly by a single structural shock η , then the observed spillover from j to i solely reflects the propagation of η from j to i . Building on this insight, we decompose the D&Y adjacency matrix (an $N \times N$ object of normalized GFEVDs) into N shock-specific $N \times N$ matrices, each tracing spillovers attributable to a single structural shock. This decomposition yields the structural shock spillover network, which preserves the topology of the original D&Y network while restoring structural interpretation.

An immediate application of the structural shock spillover network framework is to identify “core nodes” that play pivotal roles in facilitating the spillovers of specific structural shocks throughout the network¹. This has important implications for understanding systemic vulnerability and designing targeted policy interventions. In the empirical exercise, we apply the proposed framework to China’s interbank market, aiming to analyze the spillover effects of monetary policy shocks and to identify the core interest rates. In recent years, the People’s Bank of China (PBoC) has progressively shifted its operational emphasis from quantity-oriented tools toward price-based instruments. A cornerstone of this reform has been the development of market-based benchmark rates. Nevertheless, the PBoC has not formally announced a single, well-defined operating target, leading to a multiplicity of potential policy signals.² Consequently, the transmission of monetary policy actions to the real economy and the broader financial system occurs through complex and often opaque channels (Li et al., 2021). This complexity obscures the precise mechanisms through which policy impulses propagate and complicates the measurement of the monetary policy stance,

¹The term “core nodes” is not a standard phrase. In the SVAR literature, it is well supported to treat variables that have large and/or persistent IRFs as evidence that they are central in transmitting a given shock. For example, Bernanke et al. (2005) and Gertler and Karadi (2015) highlight large and persistent responses of financial variables (e.g., credit spreads) to monetary policy shocks, effectively treating them as systemically important variables in monetary transmission. In the connectedness literature, Diebold and Yilmaz (2012) define systemically important node j as the one has large total directional connectedness to others from j (see Subsection 2.1 for detailed definition of total directional connectedness metrics).

²Chen et al. (2018) note that “For China, the intermediate target of monetary policy has been M2 growth since 2000” and argue that M2 growth targets are formally announced by the Premier of the State Council as part of the Annual Report on the Work of Government (RWG) presented to the annual session of the National People’s Congress. However, since 2018, the RWG no longer publishes M2 growth targets.

thereby underscoring the critical need to identify the core interest rates.

Specifically, we first estimate a VAR for China’s interbank rates, including pledged repo rates, interbank offered rates, and government bond yields, and identify monetary policy shocks using the high-frequency heteroskedasticity method ([Wright, 2012](#)). We then construct the monetary-policy-shock spillover network from the VAR coefficients and the identified shocks. The topology of the resulting spillover network reveals a distinctive structure: Treasury bond yields, particularly the 10-year maturity, exhibiting the strongest spillover effects and serve as the core node within the network. In contrast, money-market rates, such as repo and interbank offered rates, occupy more peripheral positions. This finding, derived from network analysis, presents a notable contrast to the results obtained from traditional IRF analysis, which demonstrates that money-market rates exhibit large and persistent responses to monetary policy shocks, underscoring their role as systemically important variables in the monetary policy shock transmission mechanism.

To validate these findings, we extend the multi-step regression approach of [Altavilla et al. \(2017\)](#) to assess how shock series, constructed from interest rate innovations and impact coefficients derived from the previously estimated SVAR, improve the goodness-of-fit in predicting the future interest rate across multiple horizons compared to a historical-information benchmark. The results are clear: shocks identified from Treasury bond yields explain variation across the broader interest rate spectrum more effectively than those derived from money-market rates. This outcome demonstrates a clear alignment between network topology and fitting performance in that topologically important nodes deliver the most accurate characterization of monetary policy transmission across the benchmark interest rate system.

Our empirical findings yield substantive implications for both policymakers and academic researchers. For policymakers, these results highlight the pivotal role of government bond yields within China’s current monetary framework, suggesting that money-market rates alone may offer an incomplete representation of the policy stance. For researchers operating in the

absence of an explicit central-bank operating target in China, using Treasury bond yields significantly enhances the identification of monetary policy shocks and the quantification of their macroeconomic and financial effects.³

This study also contributes to the networks literature. Since the global financial crisis, network models have become central to systemic-risk analysis, and the D&Y connectedness framework has been especially influential for its scalability and granularity (Diebold and Yilmaz, 2023). We offer a structural interpretation of D&Y edges by emphasizing the role of structural shocks in shaping network connections. Recent studies similarly highlight the importance of common shocks in generating interdependence: whereas much of the contagion literature focuses on spillovers from idiosyncratic shocks, comovement among financial variables frequently reflects common (macroeconomic) disturbances. In response, several papers seek to partial out common components to measure idiosyncratic contagion more cleanly.⁴ Our focus is different. Although the structural shocks we study are a form of common shock, we analyze how these shocks generate spillover networks and determine their topology. Methodologically, we depart from related approaches by working within a VAR-based framework and imposing identified shocks to yield a structural decomposition of connectedness. Thus, our extension preserves the original D&Y construction while supplying a structural interpretation of its edges—clarifying the economic content of connectedness and providing a foundation for shock-specific analysis in macro-financial settings.

The literature most closely related to ours is Santis and Zimic (2018). Santis and Zimic (2018) first identify a full set of structural shocks and then use structural FEVDs to construct their networks. They use their network to locate the nodes that serve as primary sources of

³For instance, event-study methodologies commonly require a representative proxy for the monetary policy instrument. While Sun (2020) note that the 7-day interbank rate may be foremost among those candidates and studies such as Das and Song (2023) and Shieh (2024) consequently adopt swaps based on 7-day repo rate as their policy instrument, our analysis reveals that repo rate is not a core node within the monetary spillover network and conveys substantially less information about China’s monetary policy than government bond yields.

⁴Barigozzi and Hallin (2016) and Barigozzi et al. (2021) use a General Dynamic Factor Model to separate common and idiosyncratic shocks, while Ando et al. (2022) incorporate observable common factors as exogenous controls in the VAR to identify idiosyncratic shocks.

spillovers. However, in their framework, each node is subject to influence from all structural shocks in the system. Thus, their “main nodes” reflect the combined effects of every specified shock rather than the distinctive role of any single structural driver. By contrast, our approach attributes spillovers at each node to a single structural shock and permits partial identification of a subset of shocks. In our empirical application, for instance, we focus exclusively on the monetary policy shock. This focused attribution enables the construction of a structural shock spillover network that is shock-specific, thereby offering clearer economic interpretation and more directly informing shock-specific policy analyses.

The remainder of this paper is organized as follows. Section 2 introduces the theoretical foundation and construction of the structural shock spillover network. Section 3 describes the empirical procedures and data. Section 4 presents the empirical results. Section 5 validates these results using the multi-step regression strategy of [Altavilla et al. \(2017\)](#). Section 6 conclusions.

2 The Structural Spillover Network

In this section, we introduce the structural spillover network, which is an extension of the D&Y network integrated with the SVAR framework. We first review the D&Y network, then detail on how to construct the structural spillover network.

2.1 A brief review of the D&Y network

The D&Y network is a data-driven connectedness framework that builds on GFEVDs. Consider a (stationary) network consisting of N variables collected in x_t with the vector moving-average (VMA) representation:

$$x_t = \sum_{h=0}^{\infty} A_h \varepsilon_{t-h}, \quad \varepsilon_t \sim \mathcal{N}(0, \Sigma), \quad (1)$$

where $\{A_h\}_{h=0}^{\infty}$ are $N \times N$ parameter matrices, and ε_t is a $N \times 1$ vector of innovations. The GFEVD, which quantifies the extent to which information about future innovations of $x_{j,t}$ reduces the H -step-ahead MSFE of $x_{i,t}$, is defined as:

$$\theta_{i,j}^H := \frac{\mathbb{E} [x_{i,t+H} - \mathbb{E}(x_{i,t+H} | \Theta_0)]^2 - \mathbb{E} [x_{i,t+H} - \mathbb{E}(x_{i,t+H} | \Theta_0, \Theta_j)]^2}{\mathbb{E} [x_{i,t+H} - \mathbb{E}(x_{i,t+H} | \Theta_0)]^2}, \quad (2)$$

where $\Theta_0 := \{\varepsilon_{t-h}\}_{h=0}^{\infty}$ is the information set containing all the past innovations of x_t up to time t , and $\Theta_j := \{\varepsilon_{j,t+h}\}_{h=1}^H$ is the information set containing the innovations of $x_{j,t}$ from $t+1$ to $t+H$. The adjacency matrix $\mathbf{D}^H = [d_{i,j}^H]$ of the D&Y network is the normalized $\theta_{i,j}^H$:

$$d_{i,j}^H := \frac{\theta_{i,j}^H}{\sum_{j'=1}^N \theta_{i,j'}^H}, \quad (3)$$

with the normalizing condition $\sum_{j=1}^N d_{i,j}^H = 1$. Then the gross pairwise directional spillover from j to i is defined as

$$C_{i \leftarrow j}^H := d_{i,j}^H. \quad (4)$$

Building on the the gross pairwise directional spillover index, related spillover indices include the net pairwise directional spillover: $C_{i,j}^H = C_{j \leftarrow i}^H - C_{i \leftarrow j}^H$; the total directional spillover from others (to i): $C_{i \leftarrow \bullet}^H = \sum_{j=1, j \neq i}^N C_{i \leftarrow j}^H$; the total directional spillover to others (from j): $C_{\bullet \leftarrow j}^H = \sum_{i=1, i \neq j}^N C_{i \leftarrow j}^H$; the net total directional spillover of variable i : $C_i^H = C_{\bullet \leftarrow i}^H - C_{i \leftarrow \bullet}^H$; and the total spillover: $C^H = N^{-1} \sum_{i,j=1, i \neq j}^N d_{i,j}^H$.

Despite its empirical appeal, the D&Y network is a reduced-form construct: each edge aggregates the predictive content of reduced-form innovations ε_t without distinguishing their underlying structural sources. Consequently, the economic interpretation of spillovers is limited. For example, one cannot separate the portion due to monetary, growth, or risk-premium shocks (Cieslak and Pang, 2021), or conduct shock-specific policy counterfactuals. This limitation motivates the structural spillover network developed below, which embeds the D&Y framework in an SVAR and decomposes each edge into shock-specific components, thereby delivering a clear structural interpretation.

2.2 The structural spillover network

To advance from the D&Y network to the structural spillover network, the key step is to decompose the D&Y adjacency matrix (Eq.(3)) into submatrices attributable to different structural shocks. The D&Y adjacency matrix derives from the GFEVDs (Eq.(2)), which depends on the following conditional expectation:

$$\mathbb{E}(x_{i,t+H} | \Theta_0, \Theta_j) = \mathbb{E}(x_{i,t+H} | \Theta_0) + \sum_{h=0}^{H-1} e_i' A_h \mathbb{E}(\varepsilon_{t+H-h} | \varepsilon_{j,t+H-h}), \quad (5)$$

where e_i is the i -th column of the N dimensional identity matrix I_N . In Eq.(5), the additional information Θ_j brought in helping forecasting $x_{i,t}$ depends on the correlation between ε_t and $\varepsilon_{j,t}$ ⁵. In an SVAR framework, such correlation arises from underlying structural shocks, since the innovations ε_t are assumed to be linearly driven by N mutually independent structural shocks $\eta_t = (\eta_{1,t}, \dots, \eta_{k,t}, \dots, \eta_{N,t})'$:

$$\varepsilon_t = R \eta_t = \sum_{k=1}^N R_k \eta_{k,t}, \quad \eta_t \sim \mathcal{N}(0, D), \quad D = \text{diag}(\delta_k^2), \quad (6)$$

where $R_k \in \mathbb{R}^N$ captures the impact effects of $\eta_{k,t}$ on x_t .⁶ Substituting Eq.(6) into Eq.(5) yields:

$$\mathbb{E}(x_{i,t+H} | \Theta_0, \Theta_j) - \mathbb{E}(x_{i,t+H} | \Theta_0) = \sum_{h=0}^{H-1} e'_i A_h R \mathbb{E}(\eta_{t+H-h} | \varepsilon_{j,t+H-h}). \quad (7)$$

Therefore, in the SVAR framework, the additional information Θ_j brought in helping forecasting $x_{i,t}$ depends on (i) the correlation between structural shocks η_t and the innovation $\varepsilon_{j,t}$, and (ii) R , the impact effects of these inferred structural shocks on $x_{i,t}$. Thus, in the SVAR settings, the predictive content of $\varepsilon_{j,t}$ stems from the information it contains about structural shocks.

However, the predictive information encapsulated in the innovation $\varepsilon_{j,t}$ is not specific to any single structural shock. Indeed,

$$\mathbb{E}(\varepsilon_t | \varepsilon_{j,t}) = R \mathbb{E}(\eta_t | \varepsilon_{j,t}) = \sum_{k=1}^N R_k \mathbb{E}(\eta_{k,t} | \varepsilon_{j,t}), \quad (8)$$

so any conditional forecast of ε_t based on $\varepsilon_{j,t}$ inevitably aggregates the signal reflecting contributions from the entire set of structural shocks η_t ⁷. This aggregation obscures the

⁵Since $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$, we have $\mathbb{E}(\varepsilon_t | \varepsilon_{j,t}) = \mathbb{E}(\varepsilon_t) + \text{Cov}(\varepsilon_t, \varepsilon_{j,t}) \text{Var}(\varepsilon_{j,t})^{-1} (\varepsilon_{j,t} - \mathbb{E}(\varepsilon_{j,t})) = \frac{\Sigma_{ej}}{\sigma_{jj}} \varepsilon_{j,t}$, where $\sigma_{jj} = e'_j \Sigma e_j$ denotes the variance of $\varepsilon_{j,t}$.

⁶Consistent with SVAR conventions, we assume that the matrix $R = (R_1, \dots, R_N)$ is nonsingular, which allows structural shocks η_t to be identified from the innovations ε_t .

⁷Alternatively, any structural shock $\eta_{k,t}$ could, in principle, be inferred from the whole residual vector ε_t .

individual contribution of each shock to the forecast improvement. To attribute forecast revisions to a particular structural shock $\eta_{k,t}$, we therefore need a shock-specific signal that retains only the information about $\eta_{k,t}$ conveyed by $\varepsilon_{j,t}$.

To operationalize this idea, we construct $\zeta_{k\leftarrow j,t}$ as a synthetic variable whose informative component is precisely the projection of $\eta_{k,t}$ on $\varepsilon_{j,t}$:

$$\zeta_{k\leftarrow j,t} := \mathbb{E}(\eta_{k,t} \mid \varepsilon_{j,t}) + \epsilon_{k\perp j,t}, \quad (9a)$$

$$\epsilon_{k\perp j,t} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \lambda_j), \quad \lambda_j = \delta_k^2 - \frac{R_{jk}^2 \delta_k^4}{\sigma_{jj}}, \quad (9b)$$

$$\epsilon_{k\perp j,t} \perp \varepsilon_t. \quad (9c)$$

Theorem 1 demonstrates how we can use $\zeta_{k\leftarrow j,t}$ to decompose the GFEVD matrix $[\theta_{i,j}^H]$ in Eq.(2):

Theorem 1. *Given the definition in Eq.(9), $\zeta_{k\leftarrow j,t}$ has the following properties:*

1. $\text{Var}(\zeta_{k\leftarrow j,t}) = \text{Var}(\eta_{k,t})$.
2. Using $\zeta_{k\leftarrow j,t}$ to forecast ε_t , the only useful information is $\mathbb{E}(\eta_{k,t} \mid \varepsilon_{j,t})$.
3. Given the information set $\Theta_{k\leftarrow j} := \{\zeta_{k\leftarrow j,t+h}\}_{h=1}^H$, and define $\theta_{i,k\leftarrow j}^H$ as follows:

$$\theta_{i,k\leftarrow j}^H := \frac{\mathbb{E}[x_{i,t+H} - \mathbb{E}(x_{i,t+H} \mid \Theta_0)]^2 - \mathbb{E}[x_{i,t+H} - \mathbb{E}(x_{i,t+H} \mid \Theta_0, \Theta_{k\leftarrow j})]^2}{\mathbb{E}[x_{i,t+H} - \mathbb{E}(x_{i,t+H} \mid \Theta_0)]^2}, \quad (10)$$

we have

$$\theta_{i,k\leftarrow j}^H = \frac{R_{jk}^2 \delta_k^2}{\sigma_{jj}} \theta_{i,j}^H = \omega_{jk} \theta_{i,j}^H, \quad (11a)$$

$$\theta_{i,j}^H = \sum_{k=1}^N \theta_{i,k\leftarrow j}^H. \quad (11b)$$

Proof. See Appendix A.1. □

Properties 1 and 2 in Theorem 1 imply that $\zeta_{k \leftarrow j, t}$ preserves the variance of $\eta_{k, t}$ while its informational content about $\eta_{k, t}$ is entirely mediated by $\varepsilon_{j, t}$: For each pair (k, j) , we extract the part of shock $\eta_{k, t}$ that is inferred from that single innovation $\varepsilon_{j, t}$, and we deliberately remove any extra structure that shock $\eta_{k, t}$ might have due to other innovations.⁸ Property 3 first defines $\theta_{i, k \leftarrow j}^H$ as the reduction in the H -step-ahead MSFE of $x_{i, t}$ due to $\Theta_{k \leftarrow j}$. Eq.(11a) then links $\theta_{i, k \leftarrow j}^H$ to $\theta_{i, j}^H$ via the weight $\omega_{jk} = \frac{R_{jk}^2 \delta_k^2}{\sigma_{jj}}$. Finally, Eq.(11b) implies that $\theta_{i, j}^H$ can be decomposed into N submatrices $\theta_{i, k \leftarrow j}^H$, one for each structural shock.

Using Eq.(11a) and the definition of the D&Y adjacency matrix $\mathbf{D}^H = [d_{i, j}^H]$ in Eq.(3), we can decompose $d_{i, j}^H$ into N shock-specific submatrices:

$$d_{i, k \leftarrow j}^H := \frac{\theta_{i, k \leftarrow j}^H}{\sum_{j'=1}^N \theta_{i, j'}^H} = \omega_{jk} d_{i, j}^H, \quad (13a)$$

$$d_{i, j}^H = \sum_{k=1}^N d_{i, k \leftarrow j}^H. \quad (13b)$$

Eq.(13) links each shock-specific adjacency term $d_{i, k \leftarrow j}^H$ to the original D&Y link $d_{i, j}^H$ via the weight $\omega_{jk} = \frac{R_{jk}^2 \delta_k^2}{\sigma_{jj}}$. From a network perspective, ω_{jk} measures the fraction of the pairwise directional spillover from $x_{j, t}$ to $x_{i, t}$ that can be attributed to structural shock $\eta_{k, t}$: it rescales the overall $j \rightarrow i$ spillover $d_{i, j}^H$ into the component propagated specifically by shock k through node j . Two limiting cases are illustrative:

- If $\omega_{jk} = 1$, no other structural shock affects $x_{j, t}$ contemporaneously, hence all the information in $\varepsilon_{j, t}$ is attributable to $\eta_{k, t}$. In this case, the spillover from $x_{j, t}$ to $x_{i, t}$ is

⁸To see the intuition, note that $\eta_{k, t}$ can always be decomposed into its projection on $\varepsilon_{j, t}$ and the associated residual (the orthogonal projection):

$$\eta_{k, t} = \mathbb{E}(\eta_{k, t} | \varepsilon_{j, t}) + v_{k \perp j, t}, \quad (12)$$

where $v_{k \perp j, t}$ is the remaining part of $\eta_{k, t}$ that cannot be explained by $\varepsilon_{j, t}$. By construction, $v_{k \perp j, t} \perp \varepsilon_{j, t}$, but in a multivariate system it can still carry systematic dependence with other innovations $\varepsilon_{\ell, t}$, $\ell \neq j$. In contrast, $\zeta_{k \leftarrow j, t}$ replaces this residual by the white-noise term $\epsilon_{k \perp j, t}$, which is independent of the entire innovation vector ε_t and has variance chosen so that $\text{Var}(\zeta_{k \leftarrow j, t}) = \text{Var}(\eta_{k, t})$. So $\zeta_{k \leftarrow j, t}$ and $\eta_{k, t}$ share the same marginal variance and the same conditional mean given $\varepsilon_{j, t}$, but only $\zeta_{k \leftarrow j, t}$ filters out any additional information on $\eta_{k, t}$ that might be conveyed through innovations $\varepsilon_{\ell, t}$ with $\ell \neq j$. In this sense, $\zeta_{k \leftarrow j, t}$ can be interpreted as a specific signal that retains only the part of $\eta_{k, t}$ that propagated through $\varepsilon_{j, t}$, and nothing else.

entirely driven by the propagation of $\eta_{k,t}$ through $x_{j,t}$.

- If $\omega_{jk} = 0$, $\eta_{k,t}$ has no contemporaneous effect on $x_{j,t}$; shock k does not propagate through $x_{j,t}$ and therefore contributes nothing to the spillover from $x_{j,t}$ to $x_{i,t}$.

With the decomposition Eq.(13) at hand, and analogous to the D&Y adjacency matrix, we can define the adjacency matrix of the spillover network due to structural shock $\eta_{k,t}$ ($\mathbf{D}_k^H = [d_{i,k \leftarrow j}^H]$). This also defines the gross pairwise directional spillover from j to i due to structural shock $\eta_{k,t}$:

$$C_{i,k \leftarrow j}^H = d_{i,k \leftarrow j}^H . \quad (14)$$

Based on $\mathbf{D}_k^H$, we can define the all the other network topology matrices due to structural shock $\eta_{k,t}$ analogous to the D&Y network in Subsection 2.1.

The structural spillover network attributes each edge of the D&Y network to specific structural drivers. This attribution facilitates analysis of how specific shocks spill over across nodes and horizons. In the following empirical exercise, we construct a Chinese interest rate network and apply the structural spillover framework to examine the spill over of China's monetary policy shocks and to identify core nodes within the network.

3 Application: Spillover Network of China’s Monetary Policy

Since the State Council issued the *Decision on Financial System Reform* in December 1993, interest-rate liberalization in China has proceeded steadily. A central pillar of this reform has been the development of market-based benchmark rates. However, PBoC has never explicitly announced which money-market rate it targets, nor has it provided a numerical target, leading to a multiplicity of potential policy signals. This lack of transparency obscures the precise mechanisms through which policy impulses propagate, and complicates measurement of the monetary policy stance.

To address this challenge, we build an interest-rate network and apply the structural spillover framework to trace how monetary policy shocks spill over across markets. This approach yields the network’s central nodes—those rates that mediate most spillovers—which in turn indicate the most informative instruments for inferring China’s monetary policy shocks.

3.1 China’s interest rate network

In accordance with the PBoC’s 2020 white paper, *Participating in International Benchmark Interest Rate Reform and Improving China’s Benchmark Interest Rate System*, China has established a relatively well-developed benchmark interest rate system, which includes repo rates, interbank lending rates, government and policy financial bond rates, interest rate swap curves, and the Loan Prime Rate (LPR).

Accordingly, we select nodes in the interest rate network based on the framework outlined in the white paper and data availability. First, as the PBoC conducts monetary policy primarily through the banking sector, we base the interest rate network on the interbank market, excluding rates from the exchange market. Second, we exclude the LPR due to its monthly frequency, which reduces its timeliness in reflecting policy signals, and its limited

data availability since 2019. Additionally, we exclude interest rate swap curves because they are derivative instruments dependent on underlying rates, and their inclusion might introduce informational redundancy.

We ultimately select the following rates as nodes: interbank repo rates, interbank lending rates, and interbank bond market rates. For the interbank repo market, we select the most actively traded instruments—the overnight and 7-day pledged repurchase agreements—denoted $R001$ and $R007$.⁹ For the interbank lending market, we include the overnight and one-week Shanghai Interbank Offered Rates (SHIBOR), denoted SO/N and $S1W$. For the interbank bond market, we select 3-month (TB3M), 1-year (TB1Y), 5-year (TB5Y), and 10-year (TB10Y) Treasury bond yields, representing short-, medium-, and long-term maturities.

In summary, the network comprises eight nodes: $x_t = (R001, R007, SO/N, S1W, TB3M, TB1Y, TB5Y, TB10Y)'$. The sample runs from October 8, 2006—the earliest date when all series are available—through December 31, 2023. All data are from the Wind database.

3.2 China’s monetary-policy-shock spillover network

To construct the monetary-policy-shock spillover network, we first estimate the following VAR(p) model¹⁰

$$x_t = c + \sum_{\ell=1}^p B_{\ell} x_{t-\ell} + \varepsilon_t, \quad (15)$$

and then identify the monetary policy shock in the SVAR system in Eq.(6) using the high-frequency heteroskedasticity method (Wright, 2012). Without loss of generality, we label the monetary policy shock $\eta_{1,t}$ as the first one in η_t , so identifying $\eta_{1,t}$ amounts to finding R_1 .

To implement the heteroskedasticity method, we partition the full sample $\mathcal{T} = \{1, \dots, T\}$

⁹The Depository-Institutions Repo Rate (DR), introduced in December 2014, is promoted by the PBoC as a future benchmark. DR represents the weighted-average rate of bond-collateralized repos among depository institutions, and is widely viewed as a barometer of banking system liquidity. Due to its short sample coverage, we use R rates as the repo benchmark. Although methodologically, DR and R rates are highly correlated.

¹⁰The lag length is selected by minimizing the Bayes Information Criterion (BIC), yielding $p = 2$.

into monetary policy announcement days ($\mathcal{T}_1$), when the PBoC releases policy communications, and non-announcement days ($\mathcal{T}_0$).¹¹ The identifying assumptions are: (i) the variance of the monetary policy shock is higher on $\mathcal{T}_1$ than $\mathcal{T}_0$:

$$\text{Var}(\eta_{1,t}) = \delta_{1,s}^2, \quad t \in \mathcal{T}_s, \quad s \in \{0, 1\}, \quad \delta_{1,1}^2 - \delta_{1,0}^2 > 0, \quad (16)$$

and (ii) all the other $N-1$ shocks are identically distributed over $\mathcal{T}$. Normalizing $\delta_{1,1}^2 - \delta_{1,0}^2 = 1$, the two identification assumptions imply the following moment condition for R_1 :

$$\Delta \Sigma := \Sigma_1 - \Sigma_0 = R_1 R_1', \quad (17)$$

where $\Sigma_s = \mathbb{E}[\varepsilon_t \varepsilon_t']$, $t \in \mathcal{T}_s$, $s \in \{0, 1\}$.¹² Following [Wright \(2012\)](#), we estimate R_1 using generalized method of moments (GMM):

$$\hat{R}_1 = \arg \min_{R_1} [\text{vech}(\Delta \hat{\Sigma} - R_1 R_1')]' [\text{diag}(\hat{V}_0 + \hat{V}_1)]^{-1} [\text{vech}(\Delta \hat{\Sigma} - R_1 R_1')], \quad (18)$$

where $\text{vech}(\cdot)$ is the half-vectorized operator that vectorizes the lower triangular part of a matrix, and $\hat{V}_s$ are the estimates of the variance-covariance matrices of $\text{vech}(\hat{\Sigma}_s)$, $s \in \{0, 1\}$. Given $\hat{R}_1$, the monetary policy shock series can be constructed through ([Stock and Watson, 2018](#)):

$$\hat{\eta}_{1,t} = \frac{\hat{R}_1' \text{Cov}(\hat{\varepsilon}_t)^{-1}}{\hat{R}_1' \text{Cov}(\hat{\varepsilon}_t)^{-1} \hat{R}_1} \hat{\varepsilon}_t, \quad (19)$$

¹¹The choices of the monetary policy announcement days ($\mathcal{T}_1$) and non-announcement days ($\mathcal{T}_0$) are shown in Appendix [A.2.1](#).

¹²We follow [Wright \(2012\)](#) in testing the identifying assumptions with regard to Eq.(17). The test statistics and results are shown in Appendix [A.2.2](#). Briefly speaking, all the tests are passed and the identifying assumption Eq.(17) is (statistically) valid.

with variance

$$\hat{\delta}_1^2 = \text{Cov}(\hat{\eta}_{1,t}) = (\hat{R}_1' \text{Cov}(\hat{\varepsilon}_t)^{-1} \hat{R}_1)^{-1} . \quad (20)$$

Finally, we bootstrap the VAR using the stationary bootstrap (Politis and Romano, 1994) methodology for statistical inference.

After identifying the monetary policy shock, we construct the monetary-policy-shock spillover network. We first invert the VAR in Eq.(15) to obtain the VMA in Eq.(1) and build the D&Y network $[d_{i,j}^H]$. Then, we combine the identified impact vector $\hat{R}_1$ from Eq.(18) with the full-sample shock variance (Eq.(20) with $t \in \mathcal{T}$) to compute $\omega_1 = (\omega_{11}, \omega_{21}, \dots, \omega_{N1})'$. This yields the adjacency matrix of the monetary-policy-shock spillover network, $\mathbf{D}_1^H = [d_{i,1 \leftarrow j}^H]$, and all the corresponding structural network topology matrices can then be traced out.

4 Empirical Results

4.1 The estimated $\Delta\Sigma$

We begin with the estimated $\Delta\Sigma$, the difference between the variance-covariance matrices of the VAR on $\mathcal{T}_1$ and $\mathcal{T}_0$. As shown in Eq.(17), $\Delta\Sigma$ provides the moment conditions used to identify monetary policy shocks. The estimates are reported in Table 1.

Table 1: The estimated variance-covariance shift of the VAR ($\Delta\hat{\Sigma}$)

	R001	R007	SO/N	S1W	TB3M	TB1Y	TB5Y	TB10Y
R001	−0.044	-	-	-	-	-	-	-
R007	0.004	0.201	-	-	-	-	-	-
SO/N	−0.039	0.004	−0.039	-	-	-	-	-
S1W	0.005	0.193	0.004	0.193	-	-	-	-
TB3M	0.005	0.015	0.005	0.014	0.004*	-	-	-
TB1Y	0.005*	0.014**	0.006**	0.014**	0.004***	0.005***	-	-
TB5Y	0.002	0.007**	0.002	0.007**	0.003***	0.004***	0.004***	-
TB10Y	0.004**	0.007***	0.004**	0.008***	0.003***	0.003***	0.003***	0.003***

Note: The numbers are the ordinary least squares (OLS) estimates. Statistical inferences are drawn from Politis and Romano (1994)’s stationary bootstrap method. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 1 shows sizable changes in the variance-covariance structure of Treasury bond yields between $\mathcal{T}_1$ and $\mathcal{T}_0$. By contrast, variance-covariance changes in the money-market rates are statistically insignificant. However, this does not imply that movements in money market rates provide no information in identifying the monetary policy shock. As shown in Table 1, statistically significant covariance shifts between money-market rates and Treasury bond yields still offer valuable identification insights. A plausible interpretation is that routine liquidity fluctuations dominate short-horizon volatility in money-market rates on both announcement and non-announcement days, inflating the noise in their own-variance estimates, whereas cross-covariances with Treasury yields are less contaminated by such liquidity noise and therefore more clearly reflect underlying policy information.

4.2 The impulse response functions

Table 2 summarizes key metrics derived from the IRFs, including the impact coefficient ($\hat{R}_1$), the peak coefficient $\max_h \text{IRF}(h)$, and the half-life h_f , defined as the largest h such that $\text{IRF}(h-1) \geq (1/2) \text{IRF}(1)$ and $\text{IRF}(h) \leq (1/2) \text{IRF}(1)$ (Steinsson, 2008).

Table 2: Metrics derived from the IRFs of the monetary policy shock

	$\hat{R}_1$		Peak IRF		h_f
	Coefficient	99% Confidence Interval	Value	Periods	
R001	0.06*	$[-0.030, 0.157]$	0.15	4	416
R007	0.16**	$[-0.002, 0.447]$	0.22	3	169
SO/N	0.07*	$[-0.038, 0.160]$	0.15	4	395
S1W	0.16***	$[0.002, 0.446]$	0.21	3	149
TB3M	0.06***	$[0.016, 0.090]$	0.10	5	510
TB1Y	0.07***	$[0.037, 0.093]$	0.10	5	448
TB5Y	0.05***	$[0.023, 0.080]$	0.08	5	377
TB10Y	0.05***	$[0.025, 0.073]$	0.07	4	288

Note: The first column shows the GMM estimates of R_1 in Eq.(18), which is the impact IRFs. Statistical inferences are drawn from Politis and Romano (1994)’s stationary bootstrap method. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. The third column is the peak value of IRFs, and the fourth column the periods when the peak IRFs occurs. The last column is the half-life of IRFs.

The first two columns of Table 2 report $\hat{R}_1$, the impact effects of the monetary policy shock estimated via GMM in Eq.(18), together with bootstrapped significance levels and 99% confidence intervals. All interest rates display a statistically significant impact response to the monetary policy shock at the 10% level.

When we use the impact IRFs to identify candidate key rates in the transmission of monetary policy shocks, R007 and S1W emerge as those with the largest initial responses. The peak IRF values, shown in the third column, are consistent with this evidence: the monetary policy shock generates the largest peak responses in R007 and S1W, both occurring three days after the shock. However, the persistence patterns, measured by the half-lives reported in the last column, tell a different story. Specifically, TB3M has the longest half-life (510 days), followed by TB1Y (448 days) and R001 (416 days), whereas R007 and S1W exhibit the shortest half-lives, 169 and 149 days, respectively.

Taken together, the IRF results show that money-market rates respond strongly to monetary policy shocks, but the magnitude and persistence of these responses differ substantially across maturities. While R007 and S1W dominate in terms of contemporaneous and peak responses, their short half-lives suggest that they mainly reflect short-run adjustments in funding conditions rather than the persistent component of policy transmission. This divergence between amplitude and persistence cautions against relying solely on IRFs to identify key policy-transmission rates and motivates complementary criteria. In the next subsection, we turn to the structural spillover network analysis. This framework evaluates how monetary policy shocks spillover across market segments and which nodes are most influential in disseminating policy information.

4.3 Spillover Network of China’s Monetary Policy

In this subsection, our analysis focuses on: (i) the pairwise directional spillover; (ii) the net total directional spillover of node i ; and (iii) the total spillover. Together, these objects reveal how China’s monetary policy shocks spill over across the interest-rate network and which rates function as core nodes.

4.3.1 The pairwise directional spillover

Figure 1 presents heat maps of the gross pairwise directional spillovers defined in Eq.(14) with $k = 1$ for horizons $H = \{1, 5, 22, 120, 250\}$ and $H = \infty$, which we call the steady-state, and is numerically approximated at a horizon of 2000 days. For each off-diagonal cell (i, j) with $i \neq j$, the value records the spillover from rate j to rate i that is attributable solely to the monetary policy shock. The color scale ranges from dark blue (lowest spillovers) to red (highest spillovers), with intermediate intensities represented by light blue, green, yellow, and orange.

We first examine spillovers within the Treasury bond and money-market subnetworks. Figure 1 shows that, at all horizons, the Treasury bond subnetwork (rows/columns 5–8) dis-

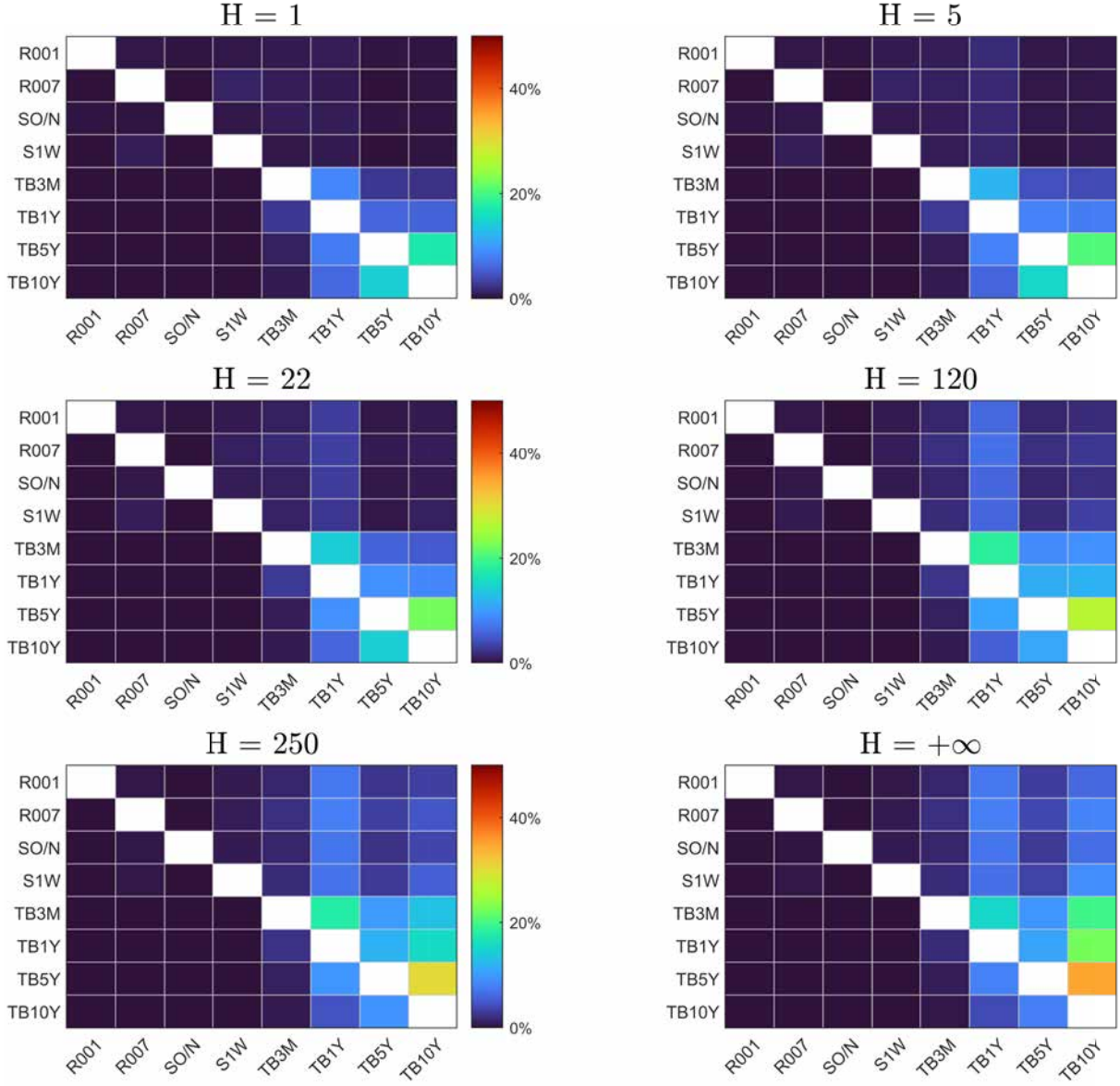


Figure 1: Heat map of the gross pairwise directional spillover metrics of the monetary-policy-shock spillover network.

Note: Each (i, j) , $i \neq j$ cell records the gross pairwise directional spillover from rate j to rate i due to the monetary policy shock. The color scale ranges from dark blue (lowest spillovers) to red (highest spillovers), with intermediate intensities represented by light blue, green, yellow, and orange.

plays systematically higher spillover intensities than the money-market subnetwork (rows/columns 1–4), indicating stronger policy-induced interactions among bond yields and much weaker spillovers among short-term money-market instruments. Within the Treasury bond subnetwork, spillovers are more pronounced from long-maturity yields (TB5Y, TB10Y; rows 5–8, columns 7–8) than from short-maturity yields (TB3M, TB1Y; rows 5–8, columns 5–6), with the exception that TB1Y generates stronger spillovers to TB3M (cell (5, 6)) than other bond maturities.

Figure 1 also reveals asymmetric spillovers cross markets. Along all horizons, Treasury bond yields exert stronger spillovers on money-market rates (rows 1–4, columns 5–8) and the spillover effects rise as the horizon lengthens, while the reverse direction remains limited (rows 5–8, columns 1–4). Within the cross-market linkages, TB1Y emerges as the dominant spillover channel, exerting significant influence on money-market rates (rows 1–4, column 6). However, at the steady state, the gross spillovers from TB10Y to money-market rates become comparable in magnitude to those from TB1Y.

Taken together, these patterns indicate that TB1Y plays a leading role in transmitting monetary policy shock effects from the bond market to short-term money-market rates, whereas TB10Y acts as the main transmitter of shocks along the medium- and long-term segments of the Treasury yield curve.

To reinforce the above finds, Figure 2 presents heat maps of the net pairwise directional spillover measure $C_{ij,1}^H = C_{j,1 \leftarrow i}^H - C_{i,1 \leftarrow j}^H$. Given that $C_{ii,1}^H = 0$ and $C_{ij,1}^H = -C_{ji,1}^H$, we report only the upper-triangular elements of $[C_{ij,1}^H]$. The patterns in Figure 2 highlight two Treasury yields as key net transmitters of monetary policy shocks. First, across all horizons, TB1Y generates stronger net spillovers to money-market rates (rows 1–4, column 6) and to the short-term TB3M (cell (5,6)) than bond yields of other maturities. Second, TB10Y consistently exerts positive net spillovers to the rest of the Treasury yield curve at all horizons; at the steady state, its net spillovers to money-market rates are comparable in magnitude to those of TB1Y.

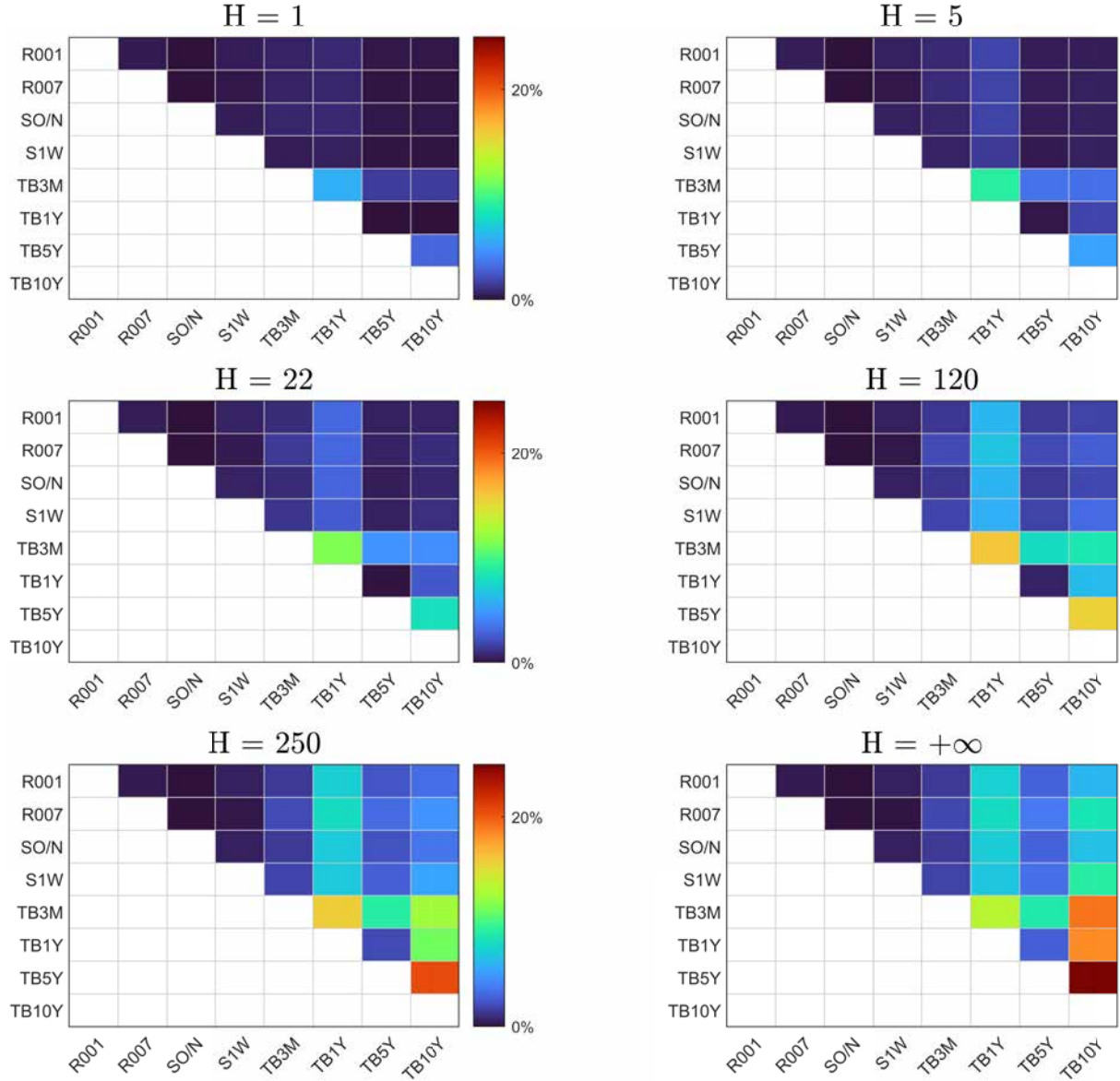


Figure 2: Heat map of the net pairwise directional spillover of the monetary-policy-shock spillover network.

Note: See Figure 1.

Overall, the network topology identifies TB1Y and TB10Y as core nodes in the monetary-policy-shock spillover network. The contrast between these network-based findings and the IRF evidence in Subsection 4.2 highlights the multi-dimensional nature of monetary policy propagation in China: while monetary policy shocks may originate or concentrate transmission in the money market, the bond market serves as a critical and responsive conduit for policy spillovers.

4.3.2 Digress: a comparison with the D&Y network

To compare the monetary-policy-shock spillover network with the D&Y network, Figure 3 shows the heat maps of the gross pairwise directional spillover metrics of the D&Y grosswork (Eq.(4)). In sharp contrast to Figure 1, Figure 3 exhibits strong spillovers within both the Treasury bond and money-market subgrossworks. Moreover, cross-market spillovers appear approximately symmetric: Treasury bond yields exert strong spillovers on money-market rates, and vice versa.

The discrepancy stems from the decomposition in Eq. (13). While Figure 1 visualizes the matrix $[d_{i,1\leftarrow j}^H]$, capturing only the policy-attributable component, Figure 3 visualizes the full spillover matrix $[d_{i,j}^H]$. The relationship $d_{i,1\leftarrow j}^H = \omega_{j1} d_{i,j}^H$ implies that the policy-shock network reflects a scaled version of the full D&Y network, with the scaling factor ω_{j1} quantifying the proportion of the spillover from $x_{j,t}$ to $x_{i,t}$ that is due to the monetary policy shock.

As argued in Subsection 4.1, routine liquidity fluctuations dominate short-horizon volatility in money-market rates, inflating noise in their own-variance estimates and diluting the share of variance explained by the policy shock. Consequently, structural shocks other than monetary policy, such as liquidity, funding, or market-microstructure shocks, play a pre-dominant role in shaping spillovers among money-market rates in the full D&Y network. This explains why the unrestricted D&Y network exhibits strong and relatively symmetric cross-market spillovers, whereas the policy-specific network reveals a more selective and asymmetric propagation pattern centered on Treasury bonds.

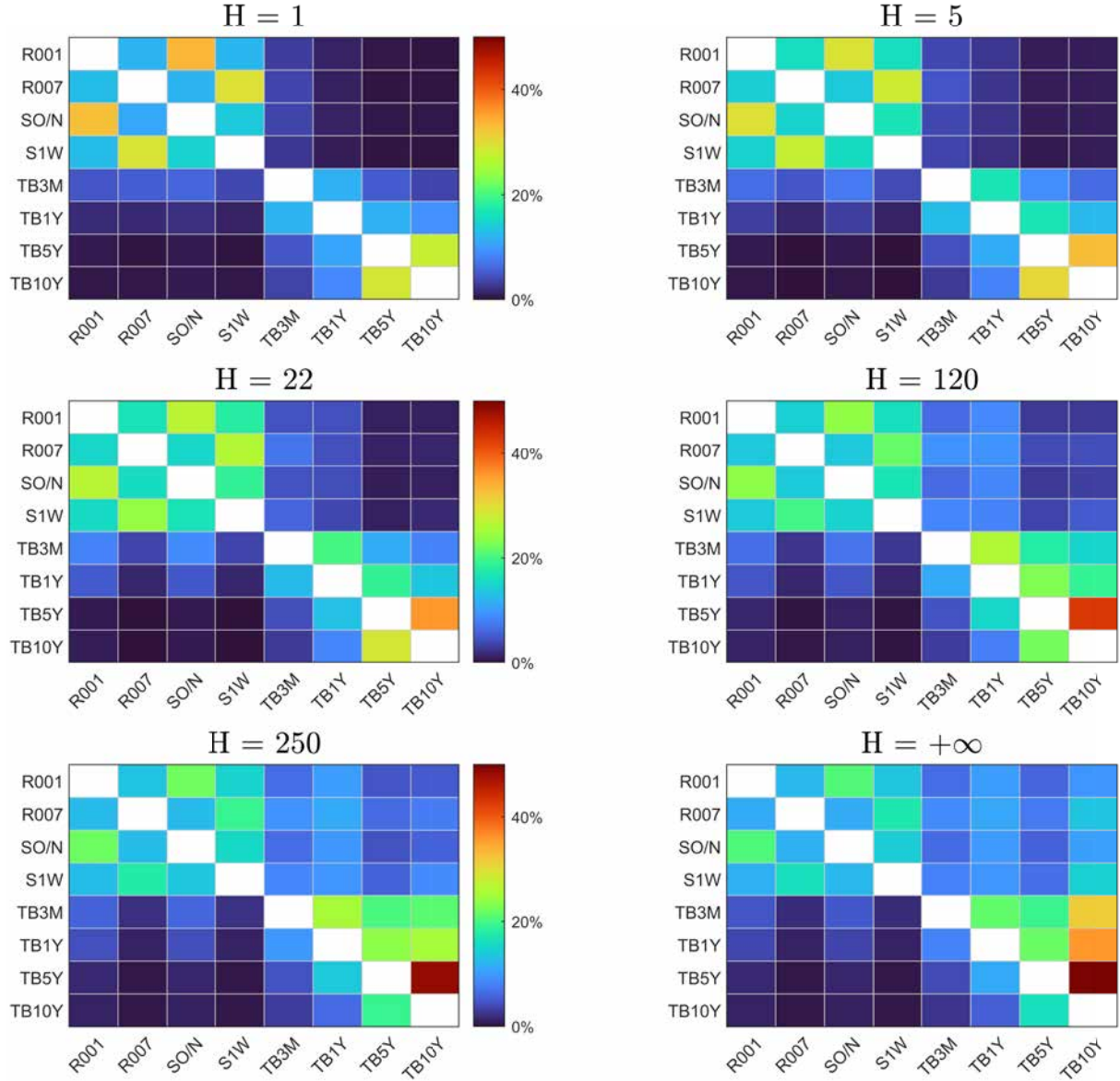


Figure 3: Heat map of the gross pairwise directional spillover metrics of the D&Y spillover network

Note: See Figure 1.

4.3.3 The net total directional spillover

Table 3 reports the net total directional spillover measure, $C_{i,1}^H = C_{\bullet,1 \leftarrow i}^H - C_{i,1 \leftarrow \bullet}^H$. A positive value indicates that node i is a net transmitter, while a negative value indicates a net receiver. Magnitudes reflect each node’s relative importance in propagating policy information through the network.

Table 3: Net total directional spillover of the monetary-policy-shock spillover network.

H	R001	R007	SO/N	S1W	TB3M	TB1Y	TB5Y	TB10Y
1	−2.66***	−1.34	−2.82***	−0.09	−6.36**	10.99*	−2.52	4.80
5	−4.55***	−2.99	−4.29***	−1.44	−12.87***	13.67*	−0.02	12.49*
22	−6.37***	−5.54*	−6.15***	−3.71	−16.19***	20.65*	−1.03	18.33**
120	−11.65***	−12.86***	−11.53***	−11.08**	−25.95***	33.64*	−0.48	39.90***
250	−15.15***	−17.33***	−15.09***	−15.62**	−30.54***	31.59	1.05	61.09***
$+\infty$	−18.56***	−21.42***	−18.52***	−19.69**	−34.79***	21.29	−2.56	94.25***

Note: The numbers are the OLS estimates. Statistical inferences are drawn from [Politis and Romano \(1994\)](#)’s stationary bootstrap method. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. The last row indicates the steady-state shares, which are numerically approximated at a horizon of 2000 days.

Across all horizons, TB1Y and TB10Y consistently function as net transmitters. All remaining nodes are net receivers, with the only exception being TB5Y, which becomes a net transmitter at the 250-day horizon. The network’s granularity also permits a market-based decomposition of each node’s net total directional spillover into its contributions to the money market and to the bond market (Appendix A.3). Table 4 shows this decomposition. As the horizon lengthens, TB1Y’s net spillover to the money market rises and eventually exceeds its net spillover to the bond market, while TB10Y’s net spillover is consistently larger toward the bond market.

Combined with Figure 1, this pattern suggests that TB1Y primarily transmits policy information toward shorter-term rates, including money-market instruments, whereas TB10Y transmits mainly within the bond market. This asymmetry highlights the dual role of the yield curve: the short end (TB1Y) bridges policy to money markets; the long end (TB10Y) propagates shocks across the Treasury term structure.

Taken together, these findings align with the broader transmission mechanism of monetary policy: short- and medium-term yields primarily channel expectations of future policy rates to money-market instruments (Campbell and Shiller, 1991; Gürkaynak et al., 2005), whereas long-term yields embed both policy expectations and term premia (Cochrane and Piazzesi, 2005; Rudebusch and Wu, 2008), shaping financial conditions across the economy.

Table 4: Decomposition of net total spillover of the monetary-policy-shock spillover network.

H	R001	R007	S O/N	S 1W	TB3M	TB1Y	TB5Y	TB10Y
To money market								
1	−0.79*	0.30	−0.67	1.15*	2.42**	2.97***	0.68**	0.85**
5	−1.04**	0.54	−0.90	1.40*	3.09**	6.60***	1.57***	2.01***
22	−1.14**	0.53	−1.01	1.62*	4.54**	11.81***	2.15***	3.27***
120	−1.01**	0.49	−0.89	1.41*	6.52***	24.57***	6.50***	9.52***
250	−0.93**	0.46	−0.82	1.28*	6.66***	28.95***	10.61***	16.97***
+∞	−0.85**	0.44	−0.75	1.17*	6.41***	29.00***	12.82***	29.95***
To bond market								
1	−1.88***	−1.64**	−2.16***	−1.25**	−8.78***	8.02	−3.20	3.96
5	−3.51***	−3.53***	−3.39***	−2.84***	−15.97***	7.07	−1.59	10.49
22	−5.23***	−6.07***	−5.14***	−5.33***	−20.73***	8.84	−3.18	15.07**
120	−10.64***	−13.34***	−10.64***	−12.49***	−32.47***	9.07	−6.98	30.38***
250	−14.23***	−17.79***	−14.27***	−16.90***	−37.20***	2.63	−9.56	44.12***
+∞	−17.70***	−21.85***	−17.77***	−20.86***	−41.21***	−7.72	−15.38	64.30***

Note: The sub-table labeled “To money market” represent the net spillover to money market. The sub-table labeled “To bond market” represent the net spillover to bond market. Appendix A.3 shows how we decompose the net total spillover.

4.3.4 The total spillover

Finally, we decompose the total spillover of monetary policy shocks C_1^H from a market perspective. Following Appendix A.3, we separate the aggregate measure into four components: (i) bond $\rightarrow$ money; (ii) money $\rightarrow$ bond; (iii) within monetary market; and (iv) within bond markets. This decomposition allows us to illustrate how monetary policy shocks propagate throughout the interest rate network via both within-market and cross-market linkages. The results are presented in Figure 4.

At all horizons, within-bond spillovers constitute the largest component. As the horizon

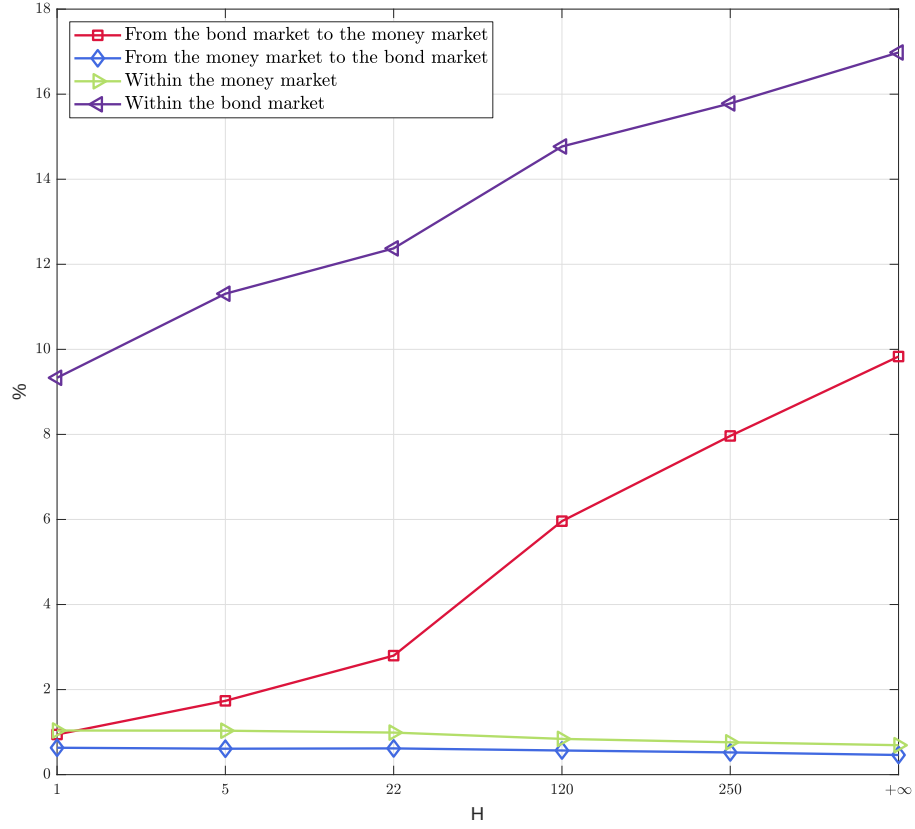


Figure 4: Decomposition of the total spillover of the monetary-policy-shock spillover network. *Note: The figure decomposes the total network spillover from a market perspective into four components: the spillover from the Treasury bond market to the money market (indicated by squares), the spillover from the money market to the Treasury bond market (indicated by diamonds), the total spillover within the money market (indicated by right triangles), and the total spillover within the government bond market (indicated by left triangles).*

increases, the bond $\rightarrow$ money component grows steadily, taking a larger share of the total. In contrast, within-money and money $\rightarrow$ bond components remain small. Hence, monetary policy shocks are transmitted primarily within the Treasury bond market and, secondarily, from the Treasury bond market to the money market.

In conclusion, Treasury bond yields, most notably the 10-year tenors, are identified as the network's central nodes. Their role is characterized by intensive spillovers within the bond market and dominant spillovers to the money market, thereby establishing them as the primary conduits for policy shock transmission. The significance of this finding lies in the fact that Treasury bond yields best encapsulate monetary policy shocks, which in turn govern the co-movement of interest rates across the entire system. The following section presents further validation of this result.

5 Fitness of interest rate variation at different frequency

In this section, we further validate our structural shock network results in a more intuitive way: Given conceptually the core nodes as the key interest rates through which monetary policy inserts its effect, shocks identified from those core nodes should provide the largest explain power for interest rate variation conditional on the historical information. We then extend the multi-stage regression strategy proposed by [Altavilla et al. \(2017\)](#) to implement this intuition.

Let $\Theta \subset x_t$ denote a subset of interest rates. Then, for $h = 0, 1, \dots, H$, we estimate the following local projections

$$x_{i,t+h} = \gamma_h + \psi_h \eta_{1,t}^\Theta + \alpha x_{i,t-1} + e_{t+h}, \quad (21)$$

and obtain the IRFs $\{\psi_h\}_{h=0}^H$ of $x_{i,t}$ to $\eta_{1,t}^\Theta$. Finally, we run the following two regressions:

$$x_{i,t+h} = \zeta_h + \beta_h \left(\sum_{i=0}^h \psi_{h-i} \eta_{1,t+i}^\Theta \right) + \rho_h x_{i,t-1} + \eta_{t+h} \quad (22)$$

$$x_{i,t+h} = \mu_h + \alpha_h x_{i,t-1} + \nu_{t+h} \quad (23)$$

and use ΔR_Θ^2 , the difference in adjusted R^2 between Eq. (22) and (23), to quantify the incremental predictive content of Θ for $x_{i,t+h}$.

Our approach differs from that of [Altavilla et al. \(2017\)](#) in three key aspects, which ensure that our strategy conceptually aligns with the approach in the structural network analysis. First, we identify monetary policy shocks using observed yields and a heteroskedasticity-based identification strategy, whereas [Altavilla et al. \(2017\)](#) relies on differences between official policy releases and survey expectations. This distinction stems from the lack of market-based expectations data for China's monetary policy. Moreover, our approach enables us to evaluate which information set yields the “good” estimation of monetary policy shocks based on how well they explain changes in yields.

Second, we scale the effects of shocks by their IRFs with respect to the corresponding interest rates. Specifically, we use the cumulative form $\sum_{i=0}^h \psi_{h-i} \eta_{1,t+i}^\Theta$ as regressor in Eq.(22). This allows us to capture the dynamic effects inherent in the SVAR and structural spillover network. By contrast, [Altavilla et al. \(2017\)](#) aggregates only the immediate effects of the identified shocks.

Third, to evaluate the information content of incoming news, we estimate the improvement in model fit conditional on historical information. Specifically, we first estimate ρ_h in Eq.(22) and assess the fit of $x_{i,t-1}$ in predicting $x_{i,t+h}$. We then examine how including the term $\sum_{i=0}^h \psi_{h-i} \eta_{1,t+i}^\Theta$ enhances the predictive accuracy. This stepwise approach reflects the notion of adjacency in a spillover network. In contrast, [Altavilla et al. \(2017\)](#) regresses cumulative shocks on cumulative changes, which effectively imposes $\rho_h = 1$ in Eq.(22). We first obtain the monetary policy shocks $\eta_{1,t}^\Theta$ as

$$\eta_{1,t}^\Theta = \frac{\widehat{R}_{\Theta,1}' \text{Cov}(\widehat{\varepsilon}_{\Theta,t})^{-1}}{\widehat{R}_{\Theta,1}' \text{Cov}(\widehat{\varepsilon}_{\Theta,t})^{-1} \widehat{R}_{\Theta,1}} \widehat{\varepsilon}_{\Theta,t}, \quad t \in \mathcal{T}_1, \quad (24)$$

where $\widehat{R}_{\Theta,1}$ and $\widehat{\varepsilon}_{\Theta,t}$ are the corresponding subsets of $\widehat{R}_1$ and $\widehat{\varepsilon}_t$. We prefer this construction to re-estimating the VAR on Θ as our goal is to isolate the inefficiency introduced by shock extraction given that the impact coefficients $\widehat{R}_1$ have already been “correctly” estimated.¹³

Guided by the network results, we consider three subsets: $\Theta_M = \{R001, R007, SO/N, S1W\}$ (money market), $\Theta_B = \{TB3M, TB1Y, TB5Y, TB10Y\}$ (bond market), and $\Theta_{B \cup M} = \Theta_B \cup \Theta_M = x_t$ (combined). If Treasury bond yields play the central role suggested by the network analysis, we expect $\Delta R_{\Theta_B}^2 > \Delta R_{\Theta_M}^2$, with $\Delta R_{\Theta_{B \cup M}}^2$ providing an upper bound.

¹³In Subsection 4.1, we show that the covariance between money market rates and Treasury bond yields helps identify the impact coefficients of monetary policy shocks on money market rates, whereas the moment conditions among money market rates alone are inconsistent with the heteroskedasticity assumption. Therefore, relying solely on money market rates may not yield the “correct” impact coefficients. In Appendix A.4, we identify the impact coefficients and shock series only using innovations of money market interest rates or Treasury bond yields, and implement the same multi-stage regression to see how the inefficient impact coefficients would effect the results. The results are as expected that the shock series from the inefficient coefficients and the money market innovations deliver a worse prediction than the one from the efficient coefficients and the money market innovation

Figure 5 plots ΔR_{Θ}^2 . For predicting money-market rates, Θ_M delivers slightly higher ΔR^2 at very short horizons (up to roughly 82 days). At longer horizons, Θ_B offer larger increments in prediction accuracy, with a max $\Delta R_{\Theta_B}^2 - \Delta R_{\Theta_M}^2$ of 0.0901 in forecasting *R007* at $h = 249$. For predicting Treasury yields, $\Delta R_{\Theta_B}^2$ exceeds $\Delta R_{\Theta_M}^2$ at all horizons, with the largest gap of 0.2924 in forecasting *TB3M* at horizon 250.

Figure 5 also shows that Θ_{BUM} performs essentially on par with Θ_B . One might expect the combined set to lie between “good” (Θ_B) and “noisy” (Θ_M) information. Instead, because monetary policy shocks explain only a small share of money-market rate variation on announcement days (Section 3.2), the combined shocks put most weight on Treasury bond yields. Indeed, the correlation between shocks from Θ_{BUM} and Θ_B is 0.9794, while the correlation with Θ_M is 0.3784. Thus, adding money-market innovations does not hurt, but using Treasury yields alone already yields high-quality policy shocks.

In summary, our analysis validates the conclusions from the network analysis: In cross-market comparisons, we find that interest rates in the Treasury bond market occupy a more central position in the monetary policy shock spillover network than those in the money market. The results in this section demonstrate that using Treasury bond yields to identify monetary policy shocks indeed leads to better predictive performance regarding movements in broader market interest rates compared to using money market rates.

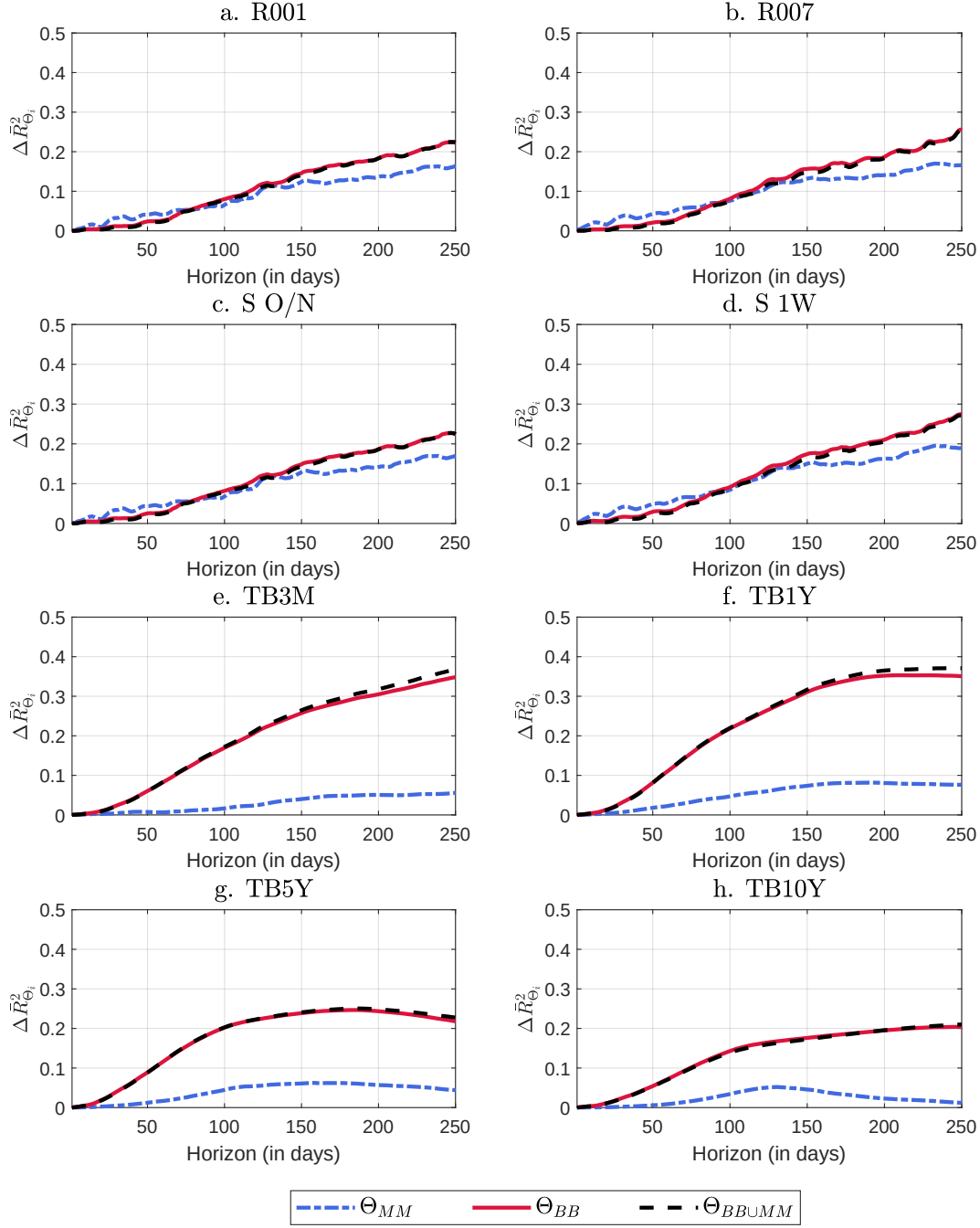


Figure 5: $\Delta \tilde{R}_{\Theta}^2$ of Θ_M , Θ_B and Θ_{BUM}

Note: The figure illustrates how information from both the Treasury bond market and money markets can enhance the accuracy of predictions. The enhanced accuracy are proxied by the difference between adjust R^2 of Eq.(22) and (23). Θ_M represents information from money markets and includes innovations of R001, R007, S O/N and S 1W from the VAR process estimated in 3.2 and Θ_B represent information from Treasury bond market and includes innovation of TB3M, TB1Y, TB5Y and TB10Y. Θ_{BUM} is the union of Θ_B and Θ_M . The coefficients used to calculate the shocks are from Table 2.

6 Conclusion

This paper proposes a structural shock spillover network that integrates the [Diebold and Yilmaz \(2012, 2014\)](#) connectedness framework into the SVAR setting, enabling us to trace how a specific structural shock propagates through the system and to identify which variables act as primary transmission nodes. Empirically, we apply the method to China’s interbank interest-rate network to study the transmission of monetary policy shocks. The results show that Treasury bond yields, particularly the 1-year and 10-year maturities, serve as core hubs through which monetary policy shocks propagate across the network. These findings are strongly validated via multi-step regressions: monetary policy shocks identified using Treasury bond yields outperform those identified with money-market rates in fitting the dynamics of interest rates. This consistency between network topology and fitting performance underscores the practical value of our structural shock spillover approach in guiding shock identification and clarifying policy transmission mechanisms.

Looking ahead, this research opens several promising avenues. First, given the ongoing improvements in China’s benchmark interest rate system, the methodology proposed in this paper can be continuously applied to identify core interest rates within the benchmark rate network. For instance, as the DR series of interbank pledged repo rates – currently being cultivated by the PBoC as short-term market benchmarks – mature and cover more extended sample periods, our framework can be used to evaluate their role in monetary policy transmission. Second, the spillover network developed here can be extended to analyze the transmission of other types of shocks, such as trade-related shocks or financial shocks. Finally, a key challenge for future work will be to integrate the network modeling approach for structural shocks (i.e., common shocks) developed in this study into other statistically based network models, so as to analyze the role of structural shocks in the formation of different types of financial or economic networks.

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A Appendix

A.1 Proof of Theorem 1

We prove a series of lemmas to help prove Theorem 1.

Lemma 1. $(\varepsilon'_t, \eta'_t)'$ are jointly normal

Proof. Since $\varepsilon_t = R \eta_t$ and $\eta_t \sim \mathcal{N}(0, D)$, $D = \text{diag}(\delta_k^2)$, we have

$$\begin{bmatrix} \varepsilon_t \\ \eta_t \end{bmatrix} = \begin{bmatrix} R \eta_t \\ \eta_t \end{bmatrix} = \begin{bmatrix} R \\ I_N \end{bmatrix} \eta_t \Rightarrow \begin{bmatrix} \varepsilon_t \\ \eta_t \end{bmatrix} \sim \mathcal{N} \left(0, \begin{bmatrix} R D R' & R D \\ D R' & D \end{bmatrix} \right)$$

□

Lemma 2.

$$\mathbb{E}(\eta_{k,t} | \varepsilon_{j,t}) = \frac{R_{jk} \delta_k^2}{\sigma_{jj}} \varepsilon_{j,t} = \alpha_{jk} \varepsilon_{j,t} \quad (\text{A.1})$$

Proof. Given the results of Lemma 1, we have

$$\mathbb{E}(\eta_{k,t} | \varepsilon_{j,t}) = \mathbb{E}(\eta_{k,t}) + \frac{\text{Cov}(\eta_{k,t}, \varepsilon_{j,t})}{\text{Var}(\varepsilon_{j,t})} (\varepsilon_{j,t} - \mathbb{E}(\varepsilon_{j,t})) = \frac{R_{jk} \delta_k^2}{\sigma_{jj}} \varepsilon_{j,t}$$

Since $\mathbb{E}(\eta_{k,t}) = \mathbb{E}(\varepsilon_{j,t}) = 0$, and $\text{Cov}(\eta_{k,t}, \varepsilon_{j,t}) = e'_k \text{Cov}(\eta_t, \varepsilon'_t) e_j = e'_k D R' e_j = R_{jk} \delta_k^2$. □

Lemma 3. $(\varepsilon'_t, \zeta'_{k \leftarrow j,t})'$ are jointly normal

Proof. According to the definition of $\zeta_{k \leftarrow j,t}$ in Eq.(9a) and the results of Lemma 2,

$$\zeta_{k \leftarrow j,t} = \mathbb{E}(\eta_{k,t} | \varepsilon_{j,t}) + \epsilon_{k \perp j,t} = \alpha_{jk} \varepsilon_{j,t} + \epsilon_{k \perp j,t} = \alpha_{jk} e'_j \varepsilon_t + \epsilon_{k \perp j,t}$$

Therefore,

$$\begin{bmatrix} \varepsilon_t \\ \zeta_{k \leftarrow j, t} \end{bmatrix} = \begin{bmatrix} \mathbf{I}_N & 0 \\ \alpha_{jk} e'_j & 1 \end{bmatrix} \begin{bmatrix} \varepsilon_t \\ \epsilon_{k \perp j, t} \end{bmatrix} = L \begin{bmatrix} \varepsilon_t \\ \epsilon_{k \perp j, t} \end{bmatrix}$$

Since $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$, $\epsilon_{k \perp j, t} \sim \mathcal{N}(0, \lambda_j)$, and $\varepsilon_t \perp \epsilon_{k \leftarrow j, t}$ (Eq.(9c)), $(\varepsilon'_t, \epsilon_{k \perp j, t})'$ are jointly normal:

$$\begin{bmatrix} \varepsilon_t \\ \epsilon_{k \perp j, t} \end{bmatrix} \sim \mathcal{N} \left(0, \begin{bmatrix} \Sigma & 0 \\ 0 & \lambda_j \end{bmatrix} \right)$$

Finally,

$$\begin{bmatrix} \varepsilon_t \\ \zeta_{k \leftarrow j, t} \end{bmatrix} \sim \mathcal{N} \left(0, L \begin{bmatrix} \Sigma & 0 \\ 0 & \lambda_j \end{bmatrix} L' \right) = \mathcal{N} \left(0, \begin{bmatrix} \Sigma & \Sigma e_j \alpha_{jk} \\ \alpha_{jk} e'_j \Sigma & \lambda_j \end{bmatrix} \right)$$

□

Lemma 4.

$$\text{Var}(\zeta_{k \leftarrow j, t}) = \delta_k^2 \tag{A.2}$$

Proof. Since $\epsilon_{k \perp j, t} \perp \varepsilon_t$ (Eq.(9c)), $\epsilon_{k \perp j, t} \perp e_j \varepsilon_t \Rightarrow \epsilon_{k \perp j, t} \perp \varepsilon_{j, t}$, and $\mathbb{E}(\eta_{k, t} | \varepsilon_{j, t}) = \alpha_{jk} \varepsilon_{j, t}$ (Lemma 2), we have $\epsilon_{k \perp j, t} \perp \mathbb{E}(\eta_{k, t} | \varepsilon_{j, t})$. Then,

$$\begin{aligned} \zeta_{k \leftarrow j, t} &= \mathbb{E}(\eta_{k, t} | \varepsilon_{j, t}) + \epsilon_{k \perp j, t} \\ \Rightarrow \text{Var}(\zeta_{k \leftarrow j, t}) &= \text{Var}(\mathbb{E}(\eta_{k, t} | \varepsilon_{j, t})) + \text{Var}(\epsilon_{k \perp j, t}) \\ &= \alpha_{jk}^2 \sigma_{jj} + \lambda_j = \frac{R_{jk}^2 \delta_k^4}{\sigma_{jj}^2} \sigma_{jj} + \delta_k^2 - \frac{R_{jk}^2 \delta_k^4}{\sigma_{jj}} = \delta_k^2 \end{aligned}$$

□

Lemma 5.

$$\mathbb{E}(\varepsilon_t | \eta_{k \leftarrow j, t}) = \sigma_{jj}^{-1} \Sigma e_j R_{jk} \eta_{k \leftarrow j, t} \quad (\text{A.3})$$

$$\text{Var}(\varepsilon_t | \eta_{k \leftarrow j, t}) = \Sigma - \sigma_{jj}^{-2} R_{jk}^2 \delta_k^2 \Sigma e_j e_j' \Sigma \quad (\text{A.4})$$

Proof. According to Lemma 3, $(\varepsilon'_t, \zeta_{k \leftarrow j, t})'$ are jointly normal, therefore,

$$\begin{aligned} \mathbb{E}(\varepsilon_t | \eta_{k \leftarrow j, t}) &= \text{Cov}(\varepsilon_t, \eta_{k \leftarrow j, t}) \text{Var}(\eta_{k \leftarrow j, t})^{-1} \eta_{k \leftarrow j, t} = \Sigma e_j \frac{\alpha_{jk}}{\delta_k^2} \eta_{k \leftarrow j, t} = \sigma_{jj}^{-1} \Sigma e_j R_{jk} \eta_{k \leftarrow j, t} \\ \text{Var}(\varepsilon_t | \eta_{k \leftarrow j, t}) &= \text{Var}(\varepsilon_t) - \text{Cov}(\varepsilon_t, \eta_{k \leftarrow j, t}) \text{Var}(\eta_{k \leftarrow j, t})^{-1} \text{Cov}(\eta_{k \leftarrow j, t}, \varepsilon_t) \\ &= \Sigma - \frac{\alpha_{jk}^2}{\delta_k^2} \Sigma e_j e_j' \Sigma = \Sigma - \sigma_{jj}^{-2} R_{jk}^2 \delta_k^2 \Sigma e_j e_j' \Sigma \end{aligned}$$

□

Lemma 6. Define $\xi_{t+H-h} = \varepsilon_{t+H-h} - \sigma_{jj}^{-1} \Sigma e_j R_{jk} \eta_{k \leftarrow j, t+H-h}$, then

- $\xi_{t+H-h} \perp \xi_{t+H-h'}, \forall h \neq h' \in \{0, \dots, H-1\}$
- The conditional variance of ξ_{t+H-h} :

$$\text{Var}(\xi_{t+H-h} | \Theta_0, \Theta_{k \leftarrow j}) = \text{Var}(\varepsilon_{t+H-h} | \eta_{k \leftarrow j, t+H-h}) = \Sigma - \sigma_{jj}^{-2} R_{jk}^2 \delta_k^2 \Sigma e_j e_j' \Sigma \quad (\text{A.5})$$

Proof. • $(\varepsilon'_s, \zeta_{k \leftarrow j, s})'$ are independent across times s , and $\xi_s = \varepsilon_s - \sigma_{jj}^{-1} \Sigma e_j R_{jk} \eta_{k \leftarrow j, s}$ is a measurable function of $(\varepsilon'_s, \zeta_{k \leftarrow j, s})'$ only, ξ_s and $\xi_{s'}$ are independent $\forall s, s'$, hence $\xi_{t+H-h} \perp \xi_{t+H-h'}, \forall h \neq h' \in \{0, \dots, H-1\}$

- The conditional variance of $\xi_{t+h-\tau}$:

$$\begin{aligned}
\text{Var}(\xi_{t+H-h}|\Theta_0, \Theta_{k \leftarrow j}) &= \text{Var}(\xi_{t+H-h}|\Theta_{k \leftarrow j}) \\
&= \text{Var}(\varepsilon_{t+H-h} - \sigma_{jj}^{-1} \Sigma e_j R_{jk} \eta_{k \leftarrow j, t+H-h} | \Theta_{k \leftarrow j}) \\
&= \text{Var}(\varepsilon_{t+H-h} - \sigma_{jj}^{-1} \Sigma e_j R_{jk} \eta_{k \leftarrow j, t+H-h} | \eta_{k \leftarrow j, t+H-h}) \\
&= \text{Var}(\varepsilon_{t+H-h} | \eta_{k \leftarrow j, t+H-h})
\end{aligned}$$

□

Lemma 7. *The conditional expectation and variance*

$$\mathbb{E}(x_{t+H}|\Theta_0, \Theta_{k \leftarrow j}) = \sum_{h=0}^{H-1} A_h \mathbb{E}(\varepsilon_{t+H-h} | \eta_{k \leftarrow j, t+H-h}) + \mathbb{E}(x_{t+H}|\Theta_0) \quad (\text{A.6})$$

$$\text{Var}(x_{t+H}|\Theta_0, \Theta_{k \leftarrow j}) = \sum_{h=0}^{H-1} A_h (\Sigma - \sigma_{jj}^{-2} R_{jk}^2 \delta_k^2 \Sigma e_j e_j' \Sigma) A_h' \quad (\text{A.7})$$

Proof. According to the VMA form of x_t in Eq.(1)

$$x_{t+H} = \sum_{h=0}^{\infty} A_h \varepsilon_{t+H-h} = \sum_{h=0}^{H-1} A_h \varepsilon_{t+H-h} + \sum_{h=H}^{\infty} A_h \varepsilon_{t+H-h}$$

Taking conditional expectation on both sides, the conditional expectation is:

$$\begin{aligned}
\mathbb{E}(x_{t+H}|\Theta_0, \Theta_{k \leftarrow j}) &= \mathbb{E}\left(\sum_{h=0}^{H-1} A_h \varepsilon_{t+H-h} | \Theta_0, \Theta_{k \leftarrow j}\right) + \mathbb{E}\left(\sum_{h=H}^{\infty} A_h \varepsilon_{t+H-h} | \Theta_0, \Theta_{k \leftarrow j}\right) \\
&= \sum_{h=0}^{H-1} A_h \mathbb{E}(\varepsilon_{t+H-h} | \Theta_0, \Theta_{k \leftarrow j}) + \mathbb{E}(x_{t+H}|\Theta_0)
\end{aligned}$$

Now consider $\mathbb{E}(\varepsilon_{t+H-h} | \Theta_0, \Theta_{k \leftarrow j})$. Since ε_t is white noise, future innovations are independent of the past information Θ_0 . Therefore, the conditional expectation depends only on $\Theta_{k \leftarrow j}$, i.e., $\mathbb{E}(\varepsilon_{t+H-h} | \Theta_0, \Theta_{k \leftarrow j}) = \mathbb{E}(\varepsilon_{t+H-h} | \Theta_{k \leftarrow j}) = \mathbb{E}(\varepsilon_{t+H-h} | \eta_{k \leftarrow j, t+H-h})$, $h \in$

$\{0, \dots, H-1\}$. Therefore

$$\mathbb{E}(x_{t+H} | \Theta_0, \Theta_{k \leftarrow j}) = \sum_{h=0}^{H-1} A_h \mathbb{E}(\varepsilon_{t+H-h} | \eta_{k \leftarrow j, t+H-h}) + \mathbb{E}(x_{t+H} | \Theta_0) \quad (\text{A.8})$$

Then, the forecast error is

$$\begin{aligned} x_{t+h} - \mathbb{E}(x_{t+h} | \Theta_0, \Theta_{k \leftarrow j}) &= x_{t+h} - \mathbb{E}(x_{t+H} | \Theta_0) - \sum_{h=0}^{H-1} A_h \mathbb{E}(\varepsilon_{t+H-h} | \eta_{k \leftarrow j, t+H-h}) \\ &= \sum_{h=0}^{H-1} A_h \varepsilon_{t+H-h} - \sum_{h=0}^{H-1} A_h \mathbb{E}(\varepsilon_{t+H-h} | \eta_{k \leftarrow j, t+H-h}) \\ &= \sum_{h=0}^{H-1} A_h (\varepsilon_{t+H-h} - \mathbb{E}(\varepsilon_{t+H-h} | \eta_{k \leftarrow j, t+H-h})) \\ &= \sum_{h=0}^{H-1} A_h (\varepsilon_{t+H-h} - \sigma_{jj}^{-1} \Sigma e_j R_{jk} \eta_{k \leftarrow j, t+H-h}) \\ &= \sum_{h=0}^{H-1} A_h \xi_{t+H-h} \end{aligned}$$

where the third equity uses the first conclusion in Lemma 5. Therefore,

$$\begin{aligned} \text{Var}(x_{t+h} | \Theta_0, \Theta_{k \leftarrow j}) &= \text{Var} \left(\sum_{h=0}^{H-1} A_h \xi_{t+H-h} | \Theta_0, \Theta_{k \leftarrow j} \right) \\ &= \sum_{h=0}^{H-1} \text{Var} (A_h \xi_{t+H-h} | \Theta_0, \Theta_{k \leftarrow j}) \\ &= \sum_{h=0}^{H-1} A_h \text{Var} (\varepsilon_{t+H-h} | \eta_{k \leftarrow j, t+H-h}) A_h' \\ &= \sum_{h=0}^{H-1} A_h (\Sigma - \sigma_{jj}^{-2} R_{jk}^2 \delta_k^2 \Sigma e_j e_j' \Sigma) A_h' \end{aligned}$$

where the second equity uses the first conclusion in Lemma 6, and the third and fourth equity use the second conclusion in Lemma 6. $\square$

Proof of Theorem 1.

1. Already proved as Lemma 3.

2. According to the prove process in Lemma 5,

$$\mathbb{E}(\varepsilon_t | \zeta_{k \leftarrow j, t}) = \frac{\alpha_{jk} \Sigma e_j}{\delta_k^2} \eta_{k \leftarrow j, t}$$

also according to Lemma 2, $\mathbb{E}(\eta_{k, t} | \varepsilon_{j, t}) = \alpha_{jk} \varepsilon_{j, t}$, hence

$$\text{Cov}(\varepsilon_t, \mathbb{E}(\eta_{k, t} | \varepsilon_{j, t})) = \text{Cov}(\varepsilon_t, \alpha_{jk} \varepsilon_{j, t}) = \alpha_{jk} \Sigma e_j$$

Therefore,

$$\mathbb{E}(\varepsilon_t | \zeta_{k \leftarrow j, t}) = \frac{\text{Cov}(\varepsilon_t, \mathbb{E}(\eta_{k, t} | \varepsilon_{j, t}))}{\delta_k^2} \zeta_{k \leftarrow j, t}$$

Hence, only $\mathbb{E}(\eta_{k, t} | \varepsilon_{j, t})$ affects the conditional expectation of ε_t on $\zeta_{k \leftarrow j, t}$ through its correlation with ε_t .

3. According to Pesaran and Shin (1998), the GFEVD defined in Eq.(2), is

$$\theta_{i, j}^H = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i' A_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} e_i' A_h \Sigma A_h' e_i}$$

According to Lemma 7 and the definition of $\theta_{i, k \leftarrow j}^H$ in Eq.(10),

$$\theta_{i, k \leftarrow j}^H = \frac{R_{jk}^2 \delta_k^2}{\sigma_{jj}} \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i' A_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} e_i' A_h \Sigma A_h' e_i} = \frac{R_{jk}^2 \delta_k^2}{\sigma_{jj}} \theta_{i, j}^H$$

According to the SVAR setting in Eq.(6), we have

$$\sigma_{jj} = e_i' \Sigma e_j = e_j' R D R' e_j = \sum_{k=1}^N R_{jk}^2 \delta_k^2 \Rightarrow \sum_{k=1}^N \frac{R_{jk}^2 \delta_k^2}{\sigma_{jj}} = 1$$

Finally, we get:

$$\theta_{i,j}^H = \sum_{k=1}^N \theta_{i,k \leftarrow j}^H$$

□

A.2 Monetary Policy Events and specification tests

A.2.1 Monetary Policy Events

A critical step in identify China’s monetary policy shocks through heteroscedasticity is to define monetary policy announcement days ($\mathcal{T}_1$). Our choice of $\mathcal{T}_1$ is broadly consistent with those in [Lin and Niu \(2020\)](#); [Sun \(2020\)](#) and [Shieh \(2024\)](#). Specifically, $\mathcal{T}_1$ includes (i) changes in the reserve requirement ratio, which is the primary quantitative policy tool of the PBoC (ii) changes in benchmark deposit and lending rates, which initially functioned as the main price based tool used in the early years of our sample but were phased out in the process of interest rate liberalization since 2015, and (iii) announcements on Medium-term Lending Facility (MLF) rates.¹⁴ There are 75 announcements during our sample period. Table [A.1](#) in Appendix [A.2.1](#) lists the dates and quantitative changes of the corresponding monetary policy instruments. As for $\mathcal{T}_0$, we take the five days preceding each announcement day as non-announcement observations to better capture volatility clustering around policy events.

Table A.1: Monetary Policy Events

No.	Date (Y/M/D)	RRR (Lg)	RRR (MS)	Dep. Rate	Loan Rate	MLF Rate
1	2006/11/3	+0.5	+0.5	—	—	—
2	2007/1/5	+0.5	+0.5	—	—	—
3	2007/2/16	+0.5	+0.5	—	—	—

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¹⁴On August 17, 2019, the PBoC released *Announcement No.15 [2019]*, which officially announced the reform and improvement of the Loan Prime Rate (LPR) formation mechanism. As LPR quotes are directly linked to MLF rates, the MLF has served as one of the main price-based tools thereafter.

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No.	Date (Y/M/D)	RRR (Lg)	RRR (MS)	Dep. Rate	Loan Rate	MLF Rate
4	2007/3/17	—	—	+0.27	+0.27	—
5	2007/4/5	+0.5	+0.5	—	—	—
6	2007/4/29	+0.5	+0.5	—	—	—
7	2007/5/18	+0.5	+0.5	+0.27	+0.18	—
8	2007/7/20	—	—	+0.27	+0.27	—
9	2007/7/30	+0.5	+0.5	—	—	—
10	2007/8/21	—	—	+0.27	+0.18	—
11	2007/9/6	+0.5	+0.5	—	—	—
12	2007/9/14	—	—	+0.27	+0.27	—
13	2007/10/13	+0.5	+0.5	—	—	—
14	2007/11/10	+0.5	+0.5	—	—	—
15	2007/12/8	+1.0	+1.0	—	—	—
16	2007/12/20	—	—	+0.27	+0.18	—
17	2008/1/16	+0.5	+0.5	—	—	—
18	2008/3/18	+0.5	+0.5	—	—	—
19	2008/4/16	+0.5	+0.5	—	—	—
20	2008/5/12	+0.5	+0.5	—	—	—
21	2008/6/7	+0.5	+0.5	—	—	—
22	2008/9/15	+0.5	-0.5	—	-0.27	—
23	2008/10/8	-0.5	-0.5	-0.27	-0.27	—
24	2008/10/29	—	—	-0.27	-0.27	—
25	2008/11/26	-1.0	-2.0	-1.08	-1.08	—
26	2008/12/22	-0.5	-0.5	-0.27	-0.27	—
27	2010/1/12	+0.5	—	—	—	—
28	2010/2/12	+0.5	—	—	—	—
29	2010/5/2	+0.5	—	—	—	—
30	2010/10/19	—	—	+0.25	+0.25	—
31	2010/11/10	+0.5	+0.5	—	—	—

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No.	Date (Y/M/D)	RRR (Lg)	RRR (MS)	Dep. Rate	Loan Rate	MLF Rate
32	2010/11/19	+0.5	+0.5	—	—	—
33	2010/12/10	+0.5	+0.5	—	—	—
34	2010/12/25	—	—	+0.25	+0.25	—
35	2011/1/14	+0.5	+0.5	—	—	—
36	2011/2/8	—	—	+0.25	+0.25	—
37	2011/2/18	+0.5	+0.5	—	—	—
38	2011/3/18	+0.5	+0.5	—	—	—
39	2011/4/5	—	—	+0.25	+0.25	—
40	2011/4/17	+0.5	+0.5	—	—	—
41	2011/5/12	+0.5	+0.5	—	—	—
42	2011/6/14	+0.5	+0.5	—	—	—
43	2011/7/6	—	—	+0.25	+0.25	—
44	2011/11/30	-0.5	-0.5	—	—	—
45	2012/2/18	-0.5	-0.5	—	—	—
46	2012/5/12	-0.5	-0.5	—	—	—
47	2012/6/7	—	—	-0.25	-0.25	—
48	2012/7/5	—	—	-0.25	-0.31	—
49	2014/11/21	—	—	-0.25	-0.40	—
50	2015/2/4	-0.5	-0.5	—	—	—
51	2015/2/28	—	—	-0.25	-0.25	—
52	2015/4/19	-1.0	-1.0	—	—	—
53	2015/5/10	—	—	-0.25	-0.25	—
54	2015/6/27	-0.5	-0.5	-0.25	-0.25	—
55	2015/8/25	-0.5	-0.5	-0.25	-0.25	—
56	2015/10/23	-0.5	-0.5	-0.25	-0.25	—
57	2016/2/29	-0.5	-0.5	—	—	—
58	2018/4/17	-1.0	-1.0	—	—	—
59	2018/6/24	-0.5	-0.5	—	—	—

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No.	Date (Y/M/D)	RRR (Lg)	RRR (MS)	Dep. Rate	Loan Rate	MLF Rate
60	2018/10/7	-1.0	-1.0	—	—	—
61	2019/1/4	-1.0	-1.0	—	—	—
62	2019/9/6	-0.5	-0.5	—	—	—
63	2019/11/5	—	—	—	—	1y: -0.05
64	2020/1/1	-0.5	-0.5	—	—	—
65	2020/2/17	—	—	—	—	1y: -0.10
66	2020/4/15	—	—	—	—	1y: -0.20
67	2021/7/9	-0.5	-0.5	—	—	—
68	2021/12/6	-0.5	-0.5	—	—	—
69	2022/1/17	—	—	—	—	1y: -0.10
70	2022/4/15	-0.25	-0.25	—	—	—
71	2022/8/15	—	—	—	—	1y: -0.10
72	2022/11/25	-0.25	-0.25	—	—	—
73	2023/3/17	-0.25	-0.25	—	—	—
74	2023/8/15	—	—	—	—	1y: -0.15
75	2023/9/14/	-0.25	-0.25	—	—	—

Notes:

1. **RRR**: Reserve Requirement Ratio; **Lg**: Large financial institutions; **MS**: Medium and small financial institutions
2. **Dep. Rate**: Deposit interest rate; **Loan Rate**: Loan interest rate
3. **MLF**: Medium-term Lending Facility (1y: 1-year)
4. All values represent percentage point changes; “+” = increase, “-” = decrease
5. “—” indicates no adjustment for that instrument on the date
6. MLF rate adjustments prior to 2019/8/17 have been excluded per research requirements
7. Data source: People’s Bank of China announcements (2006-2022)

A.2.2 Specification tests with regard to identification

In this appendix, we show two specification tests to verify the heteroscedasticity assumptions following Wright (2012). First, we test the null hypothesis that the variance-covariance matrices of the VAR residual ε_t in Eq.(15) do not differ between $\mathcal{T}_0$ and $\mathcal{T}_1$; that is, $\mathbf{H}_0^1 : \Sigma_0 = \Sigma_1$. Rejecting $\mathbf{H}_0^1$ verifies the heteroscedasticity of the structural shocks. This is tested via the test statistic:

$$W_1 = [\text{vech}(\widehat{\Sigma}_1 - \widehat{\Sigma}_0)]' [\text{diag}(\widehat{V}_1 + \widehat{V}_0)]^{-1} [\text{vech}(\widehat{\Sigma}_1 - \widehat{\Sigma}_0)] . \quad (\text{A.9})$$

The null hypothesis assumes equal covariances, and, thus, we set $\widehat{V}_0 = \widehat{V}_1 = \widehat{V}$, where $\widehat{V}$ is the estimates of the variance-covariance matrices of $\text{vech}(\widehat{\Sigma})$, the covariance of all residuals in the full sample. Because ε_t is not necessarily normally distributed, we use bootstrap hypothesis testing and compare W_1 to a distribution obtained from a bootstrap sample where the monetary policy announcement and non-announcement dates are randomly scattered while retaining the total number of announcement dates, and we obtain the bootstrap p value.

Second, we test the null hypothesis that the difference between the two reduced-form variance-covariance matrices Σ_0 and Σ_1 can be uniquely characterized by a single structural shock, where $\mathbf{H}_0^2 : \Sigma_1 - \Sigma_0 = R_1 R_1'$. Accepting $\mathbf{H}_0^2$ verifies that a single structural shock can drive the heteroscedasticity of the variance-covariance of the VAR. For this test, we compare the statistics,

$$W_2 = [\text{vech}(\widehat{\Sigma}_1 - \widehat{\Sigma}_0 - \widehat{R}_1 \widehat{R}_1')] [\text{diag}(\widehat{V}_1 + \widehat{V}_0)]^{-1} [\text{vech}(\widehat{\Sigma}_1 - \widehat{\Sigma}_0 - \widehat{R}_1 \widehat{R}_1')] , \quad (\text{A.10})$$

with the bootstrap distribution. Specifically, if we let $\widehat{\Sigma}_0^*$, $\widehat{\Sigma}_1^*$, $\widehat{V}_0^*$, $\widehat{V}_1^*$, and $\widehat{R}_1^*$ be the bootstrapped analogs of the original statistics. Then, the bootstrap distribution of the second

statistic is

$$W_2^* = \xi' [\text{diag}(\widehat{V}_0^* + \widehat{V}_1^*)]^{-1} \xi, \quad (\text{A.11})$$

where $\xi := \text{vech}(\widehat{\Sigma}_1^* - \widehat{\Sigma}_0^* - \widehat{R}_1^* \widehat{R}_1^{*'}) - \text{vech}(\widehat{\Sigma}_1 - \widehat{\Sigma}_0 - \widehat{R}_1 \widehat{R}_1')$.

Table A.2 presents the test statistics and the p value. The p value of the first specification test is approximate 0, leading us to reject the null hypothesis $\mathbf{H}_0^1$ and confirm the heteroscedasticity of the monetary policy shock. The p value of the second test is 0.7784, meaning we cannot reject the null hypothesis $\mathbf{H}_0^2$, and a single monetary policy shock can characterize the data effectively.

Table A.2: Specification tests

Hypothesis	Wald statistic	Bootstrap p value
$\Sigma_0 = \Sigma_1$	5.670	0
$\Sigma_1 - \Sigma_0 = R_1 R_1'$	13.472	0.784

A.3 Decomposition the net total spillover and total spillover

Denote $\Theta_M = \{R001, R007, SO/N, S1W\}$ the set of money market nodes and $\Theta_B = \{TB3M, TB1Y, TB5Y, TB10Y\}$ the set of bond market nodes. The net total directional spillover of variable i due to shock k is

$$\begin{aligned}
C_{i,k}^H &:= C_{\bullet,k \leftarrow i}^H - C_{i,k \leftarrow \bullet}^H = \sum_{j \neq i} (C_{j,k \leftarrow i}^H - C_{i,k \leftarrow j}^H) \\
&= \sum_{j \in \Theta_M, j \neq i} (C_{j,k \leftarrow i}^H - C_{i,k \leftarrow j}^H) + \sum_{j \in \Theta_B, j \neq i} (C_{j,k \leftarrow i}^H - C_{i,k \leftarrow j}^H)
\end{aligned} \quad (\text{A.12})$$

where $\sum_{j \in \Theta_M, j \neq i} (C_{j,k \leftarrow i}^H - C_{i,k \leftarrow j}^H)$ represents the net spillover of node i to the money market, and $\sum_{j \in \Theta_B, j \neq i} (C_{j,k \leftarrow i}^H - C_{i,k \leftarrow j}^H)$ represent the net spillover of node i to bond market.

Similarly, we decompose the total spillover using :

$$\begin{aligned}
C_k^H &:= \sum_{j \neq i} C_{j,k \leftarrow i}^H \\
&= \sum_{j \in \Theta_M, j \neq i} C_{j,k \leftarrow i}^H + \sum_{j \in \Theta_B, j \neq i} C_{j,k \leftarrow i}^H + \sum_{j \in \Theta_B, i \in \Theta_M} C_{j,k \leftarrow i}^H + \sum_{j \in \Theta_M, i \in \Theta_B} C_{j,k \leftarrow i}^H \quad (\text{A.13})
\end{aligned}$$

where

- $\sum_{j \in \Theta_M, j \neq i} C_{j,k \leftarrow i}^H$ represents the total spillover within the money market
- $\sum_{j \in \Theta_B, j \neq i} C_{j,k \leftarrow i}^H$ represents the total spillover within the bond market
- $\sum_{j \in \Theta_B, i \in \Theta_M} C_{j,k \leftarrow i}^H$ represents the total spillover from the money market to the bond market
- $\sum_{j \in \Theta_M, i \in \Theta_B} C_{j,k \leftarrow i}^H$ represents the total spillover from the bond market to the money market

A.4 Multi-stage regression using shock series with inefficient coefficients

The above analysis is based on impact coefficients identified using all the interest rates, it's provoking to see if we identify the impact coefficients and shock series only using innovations of money market interest rates or Treasury bond yields. In Figure A.1, we use money market interest rates or Treasury bond yields to identify the coefficients separately, and use the responding impact coefficients to calculate the shocks that used in the multi-stage regressions. We denote these two information set as Θ_M^* and Θ_B^* respectively. And the information set $\Theta_{B \cup M}$ is used as the benchmark in comparing the increased prediction accuracy.

Figure A.1 highlights a primary finding regarding the role of information quality in shock identification. When impact coefficients and monetary policy shocks are estimated using only Treasury bond yields, the forecasting accuracy for long-term bond yields shows a marginal

improvement over the benchmark set Θ_{BUM} . This result indicates that Θ_B^* outperforms Θ_B , particularly given the similar predictive performance of Θ_B and $\Theta_{M \cup B}$. Although both Θ_B and Θ_B^* rely on Treasury yields to construct shocks, Θ_B^* utilizes only information derived from the bond market in estimating the impact coefficients, which leads to improved forecast fit. A related insight emerges from the comparison between Θ_M and Θ_M^* : although not directly shown in Figure A.1, a comparison with Figure 5 reveals that Θ_M achieves better forecasting performance than Θ_M^* , especially for long-term Treasury yields. This is because Θ_M incorporates Treasury yields in estimating the impact coefficients used in shock construction, further underscoring the importance of high-quality information in the identification of monetary policy shocks.

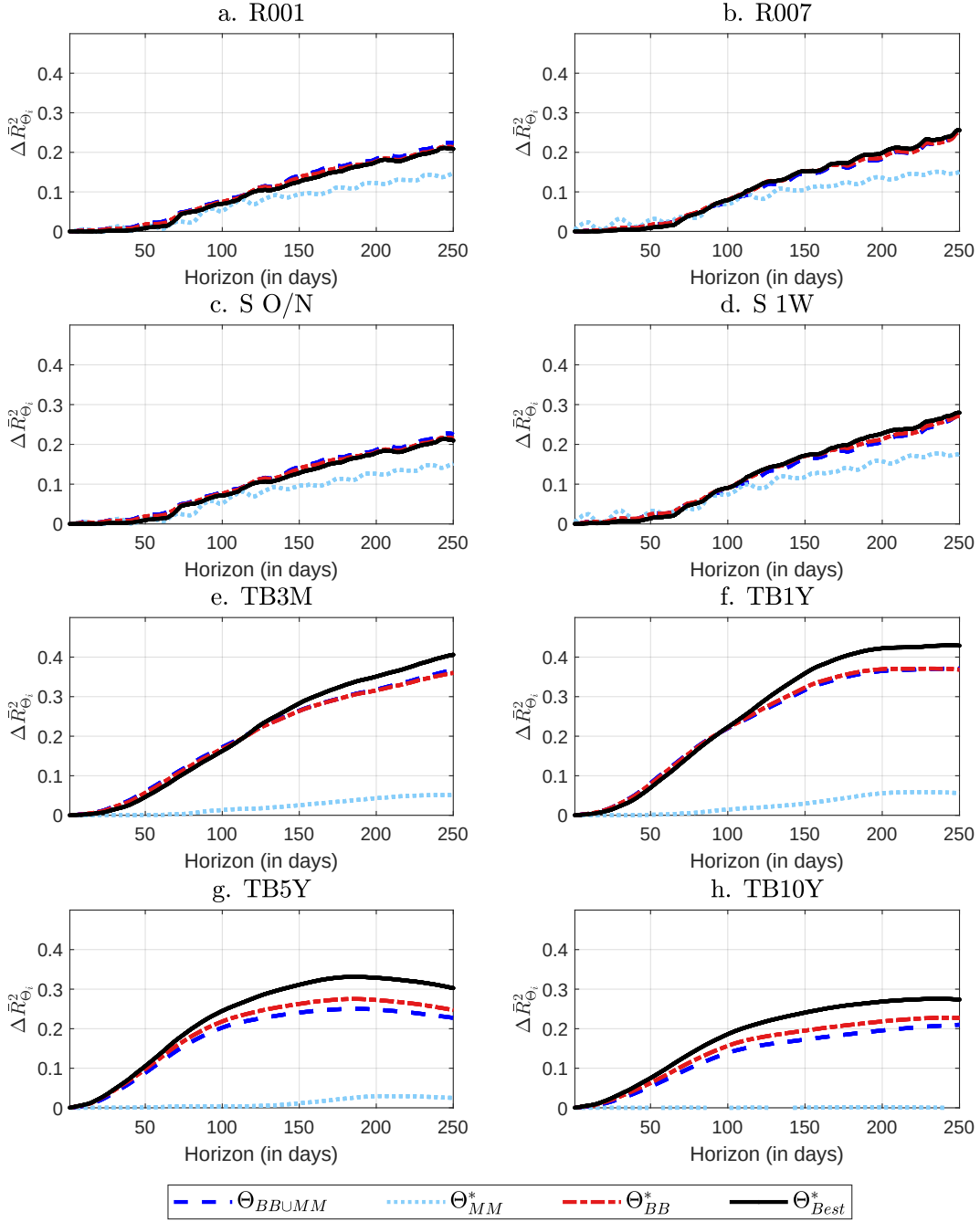


Figure A.1: $\Delta \bar{R}^2_{\Theta}$ of Θ_M^* , Θ_B^* and Θ_{BUM}

Note: This figure shows how combining information from the Treasury bond or money markets improves prediction accuracy when impact coefficients are estimated using the corresponding datasets. The improvement is measured by the difference in adjusted $\bar{R}^2$ between Equations 22 and 23. Here, Θ_M^ contains money market innovations (R001, R007, S O/N, S 1W) from the VAR in 3.2, and Θ_B^* contains bond market innovations (TB3M, TB1Y, TB5Y, TB10Y). Θ_{BUM} refers to the same information set as Figure 5. In calculating the shocks, impact coefficients used in Θ_M^* and Θ_B^* are re-estimated from each information set—not from Table 2.*